# ATYPICAL LID BUMPS

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# Lecture Acknowledgement and Thanks:

Professor William C. Hefner

Pacific University

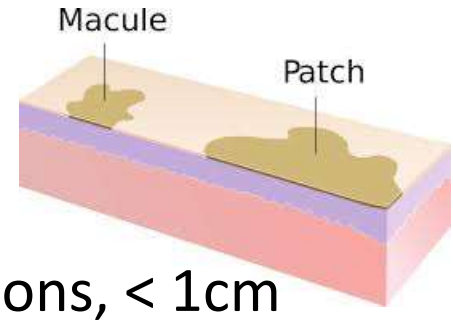


# Describe this…

- A lesion!



# Primary Dermatological Descriptors




**Macule:** Circumscribed, flat discoloration , <1cm

**Patch:** Circumscribed, flat discoloration , >1cm

**Papule:** Circumscribed, elevated superficial solid lesions, < 1cm

**Plaque:** Circumscribed, elevated superficial solid lesions, > 1cm

**Nodule:** Solid lesions with depth above, level or below surface, < 1cm

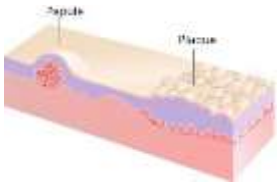
**Tumor:** Solid lesions with depth, above, level or below surface, > 1cm

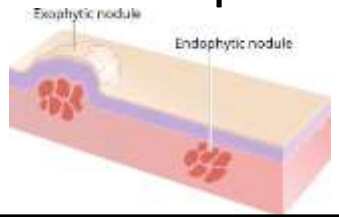
**Vesicle:** Circumscribed elevations containing serous fluid, < 1cm

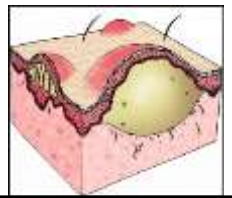
**Bulla:** Circumscribed elevations containing serous fluid, > 1cm

**Petechia:** Circumscribed deposits of blood or blood products, < 1cm

**Purpura:** Circumscribed deposits of blood or blood products, > 1cm



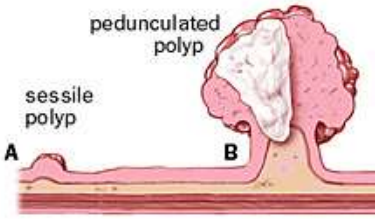


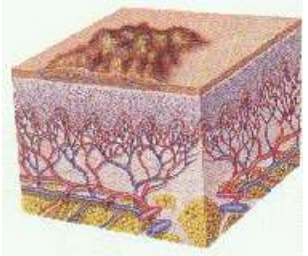



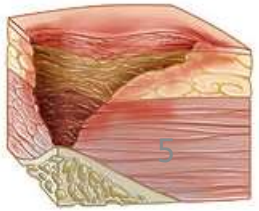


# Secondary Dermatological Descriptors

- **Sessile:** A lesion fixed to the skin on a broad base
- **Pedunculated:** A lesion on a stalk
- **Papillomatous:** A lesion exhibiting a surface resembling a cauliflower or artichoke
- **Scales:** Shedding, dead epidermal cells, dry or greasy
- **Umbilicated:** The lesion exhibits a central crater like an umbilicus or belly button
- **Crusts:** Dried masses of skin exudates
- **Ulcer:** Irregularly sized and shaped excavations extending into the dermis





# Example

- “A sessile  6mm x 10mm scaly oval nodule with ulcerative umbilicated puprurous center”



# Controlled patterns of cell growth

1. Hyperplasia
   - Increase in number of cells
2. Metaplasia
   - Change in type of adult cells which is abnormal for that tissue
3. Dysplasia
   - Alteration in size, shape, organization of adult cells

# Uncontrolled cell growth

1. Neoplasia
   - Mass of new cells which proliferate without control and serve no useful function
2. Anaplasia
   - loses resemblance to cell of origin
   - AKA tumor or cancer



# Tumors can be divided into two general classes: benign & malignant

- A malignant tumor can metastasize into lymph nodes & distant organs
- A malignant tumor, if untreated, may kill the patient wherever it occurs. Usually a benign tumor may only cause death if it happens to grow in a vital organ.
- A malignant tumor infiltrates the surrounding tissue. A benign tumor grows by expansion and is usually separated from the surrounding tissue by a capsule.
- When a malignant tumor is excised, it may recur. When a benign tumor is removed, it usually does not recur.
- Generally, a malignant tumor grows more rapidly than a benign tumor.

# Nomenclature: by parenchymal cell of origin

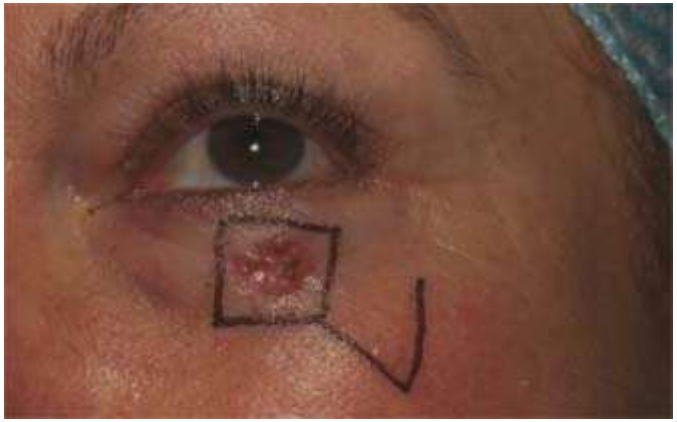
1. **Carcinoma**
   - Epithelial tissue
2. **Sarcoma**
   - Tumor of CT (derived from mesoderm), which develops into cartilage and bone (choristoma), fat (lipoma), and muscle (myoma)
3. **Hemangioma**
   - Tumor comprised of blood vessels/vasculature
4. **Lymphoma**
   - Tumor of lymphatic tissue (leukemia – hematopoietic cells affected)
5. **Melanoma**
   - Tumor of pigmented cells
6. **Hamartoma**
   - Tumor made of tissue normally present there
7. **Blastoma**
   - Tumor derived from embryonic cells (i.e. retinoblastoma)

# Tumor Treatment

1. Chemotherapy
2. Cryotherapy
3. Radiotherapy
4. Surgery







# Neoplastic considerations - Subjective

- Lesion does not act or respond as anticipated
- HX of other skin lesions elsewhere on the body
- HX of other neoplasias or systemic disease
- Pts. with HX of excessive UV skin exposure
- Lesion is not common for patient's age, sex, race, demography
- Older patients
- FHX of skin cancers
- Fair-complexion pts.
- Acute vs. chronic onset & duration
- Rapid or irregular growth patterns
- Pain or irritation associated with suspicious lesion





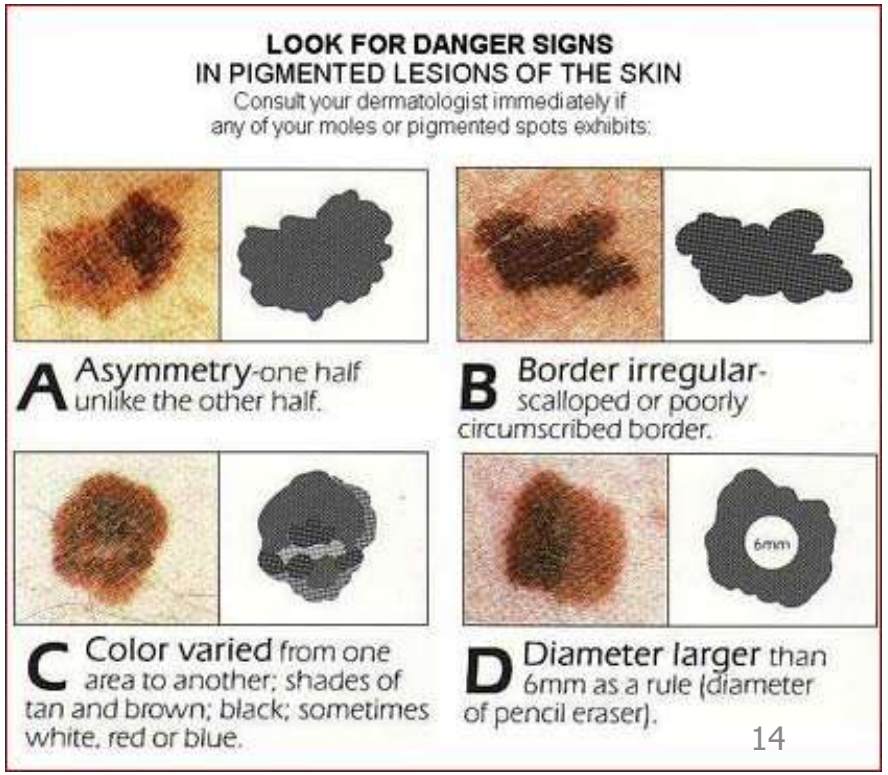
# Neoplastic considerations - Objective



- Quality of the tissue looks irregular
- Bleeding or ulceration of lesion
- Surface integrity & quality questionable
- Changes in lesion not consistent or predictable
- Neovascular patterns around or within lesion
- Recurrent infections or inflammations at site
- Uncharacteristically large lumps or bumps
- Associated tearing or conjunctival hyperemia
- Erosion of the margins or surface of a lesion

# Neoplastic Considerations - Assessment / DDX and Plan/TX

- With serious doubt or suspicion regarding any lump or bump of the lid, recommend excision and biopsy
- If referral deferred or refused for any reason, document and monitor closely
  - Photodocument if possible
  - Diagram/measure accurately
- If suspicious, RTC q. 3 mos.
- Rule of thumb :
  - **A**symmetry
  - **B**orders
  - **C**olor
  - **D**iameter
  - (**E**levation) **E**volving



# PIGMENTED LESIONS

# Ephilis



- = freckle
- Plural: ephilides
- Larger sized melanocytes (normal in number)
- Autosomal dominant inheritance pattern
- Intensified by sun exposure



# Ephelides



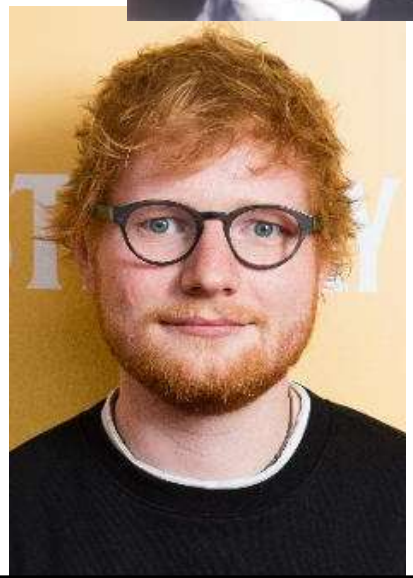


# Ephelides…
# ‘redhead’ connection?











# Solar lentigines

- = “liver spots”, age spots
- Occur in response to sunlight
- Persist in the absence of sunlight
- Middle aged to older patients
- Expanding macules (flat patches)
- Normal number of melanocytes

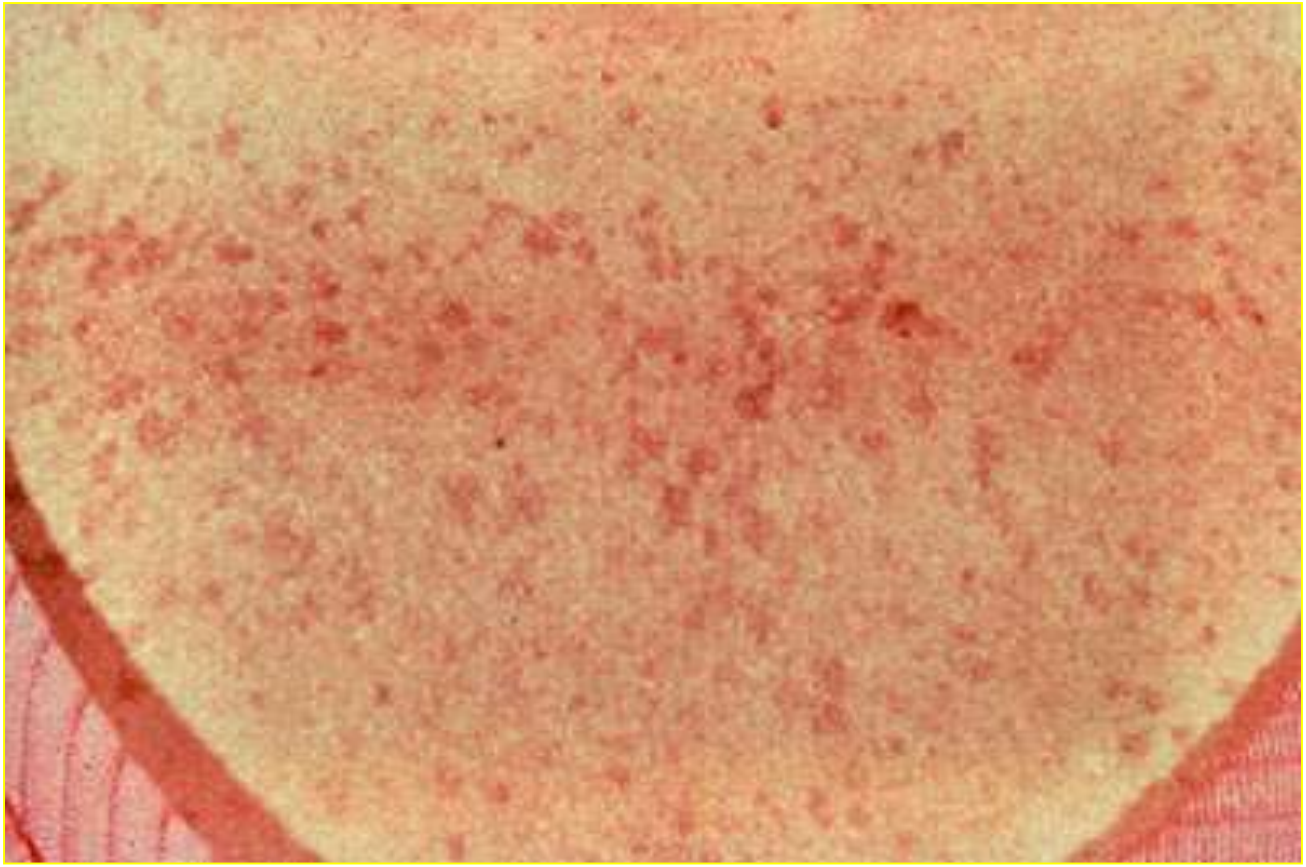









# Solar lentigines



# Solar lentigines





Pre- and post- laser treatment

# Simple lentigines



Unlike solar lentigines, simple lentigines:

1. Occur at any age
2. Have no predilection for sun-exposure or lighter skin type
3. Do not darken in response to sunlight

# Solar lentigines pre and post laser Tx





- **Mechanism:**
  Pressure waves fragment pigment particles
- **Outcome:**
  Lesions darken or grey over several days to weeks before sloughing off

# Nevus

- = mole
- Plural: nevi
- common benign neoplasms or melanocytes
- Subjective:
  - Congenital or early onset
  - Occasional changes in size or pigmentation
  - Cosmetic concern

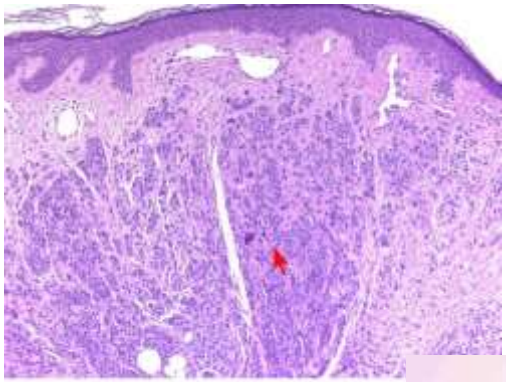




# Nevus - Objective

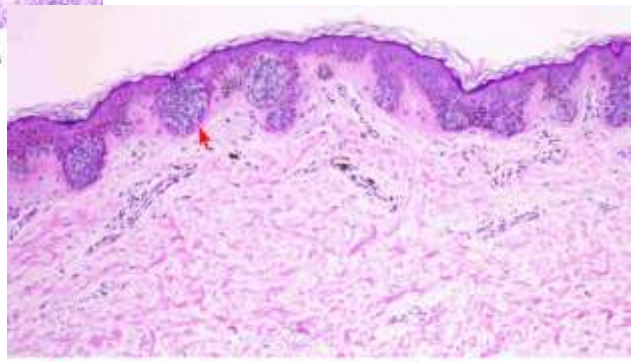
Several types:

1. Dermal
   - Most common
   - Located in dermis
   - Raised or flat



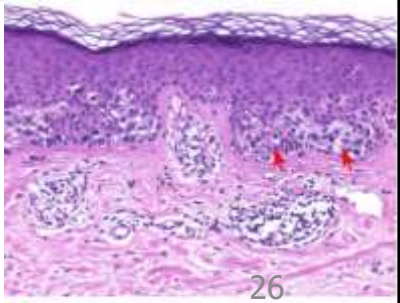
intradermal = only within the dermis

2. Junctional
   - Dermoepidermal junction
   - Superficial
   - Usually flat
   - May convert to melanoma



"nests" that are junctional (only in epidermis)

3. Compound
   - Both in dermoepidermal junction and dermis
   - Transitional



"nests" along rete ridges that are "bridged" (interconnected along their bases)

# Nevus – Dysplastic

- “Fried egg” appearance
- Increased melanoma risk if multiple and with +FHx of melanoma
- Pigmented or amelanotic
- Commonly found at lid margins
- Borders
- Less than 8 - 10 mm diameter
- May increase in size with aging
- Occasionally show hairs growing through surface

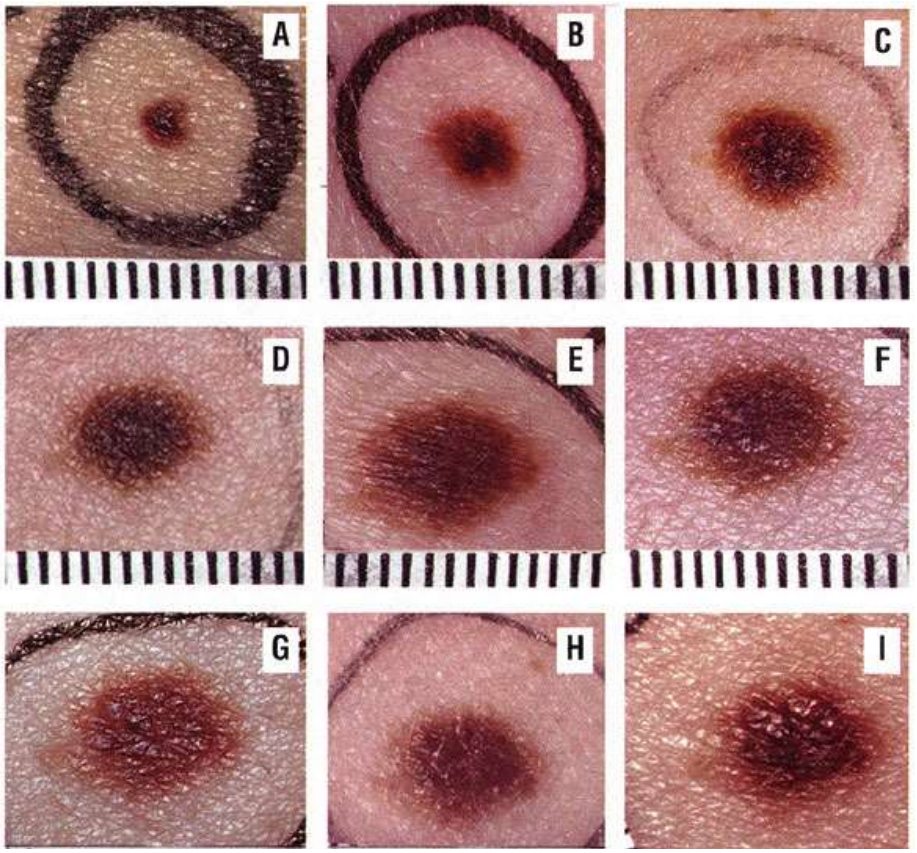


# Nevus - Assessment/DDx

- R/O other pigment spots

Plan:

- Photodocument or diagram and measure
- Monitor q. 3 - 6 months if suspicious
- RTC q. year if normal appearance with no change
- Refer for biopsy if changes occur




Dysplastic nevus change over 4 years in 26-year-old white male

# Nevus



# Dermal Nevus

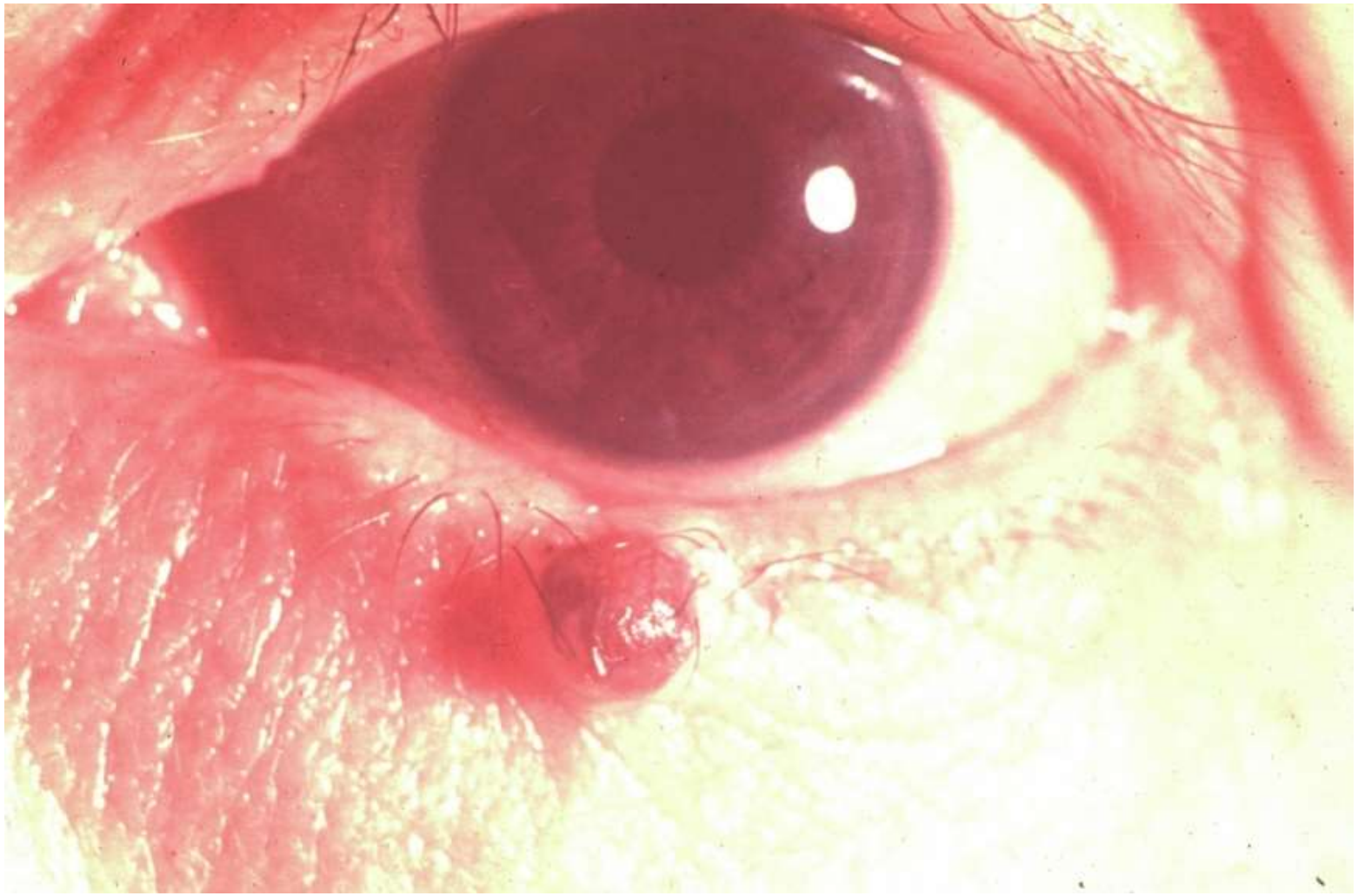


# Dermal Nevus





# Junctional Nevus

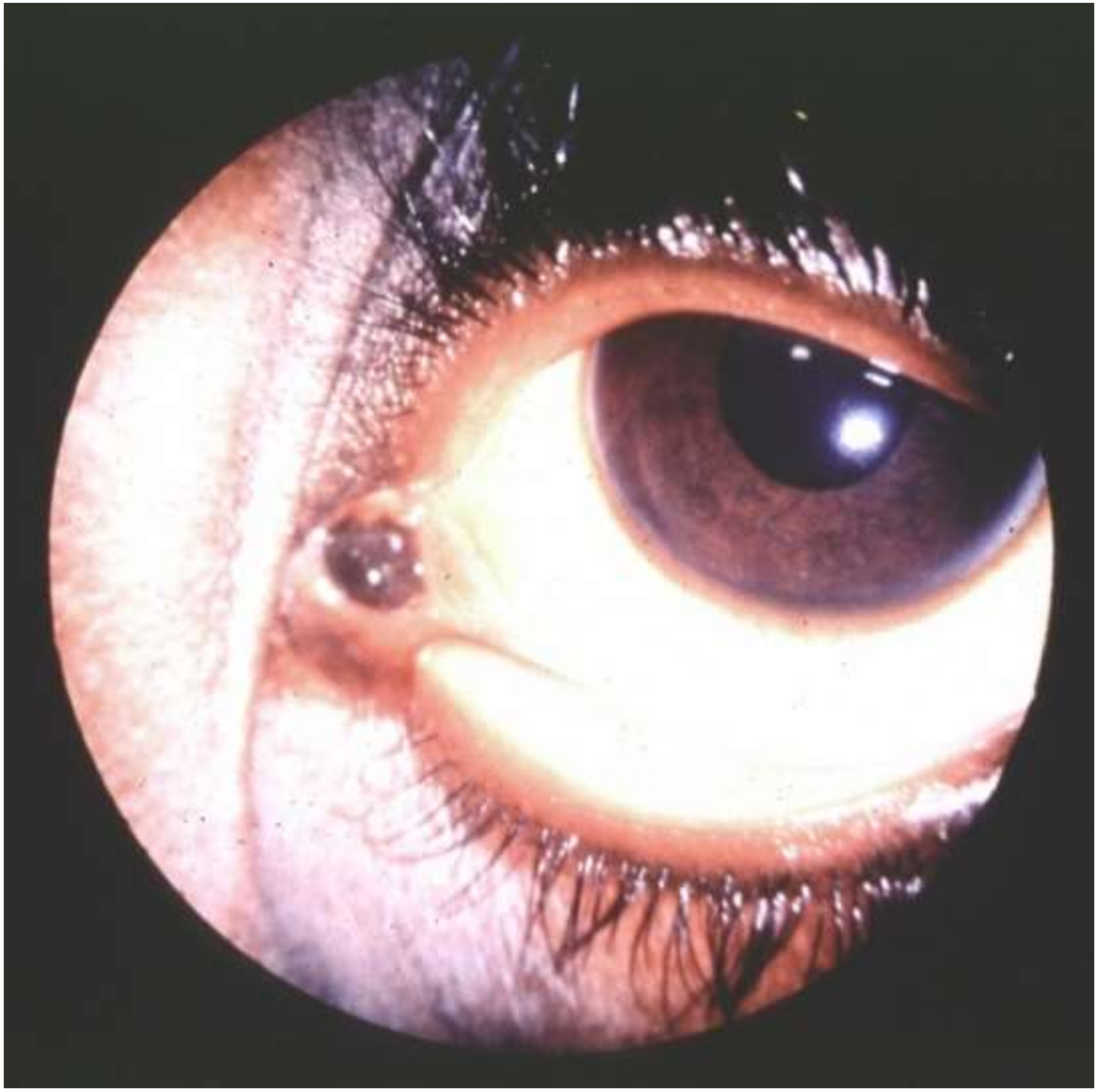


# Junctional Nevus





# Compound Nevus



# Compound Nevus





# Dysplastic Nevus







# MALIGNANT MELANOMA

# Malignant Melanoma

- Malignant tumor derived from melanocytes
- These comprise ~ 3% of primary skin cancers…
- but accounts for 79% of skin cancer related deaths!



# Malignant Melanoma - Subjective

- May not directly relate to UV exposure
- Light complexion





# Malignant Melanoma - Objective

- Asymmetric, pigmented lesion with irregular borders
- Variable coloration
- Variable diameter

Types:

1. Superficial spreading melanoma
   - Rapid increase in size
   - Lower incidence in metastases
2. Nodular melanoma
   - Vertical growth with more penetration into tissue
   - Metastasize early





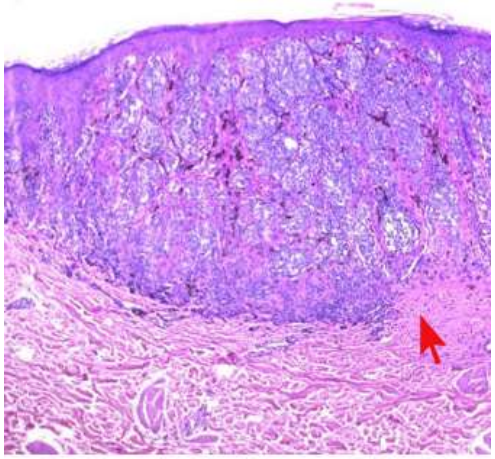

# Malignant Melanoma Mortality via Clark's levels and Breslow depth

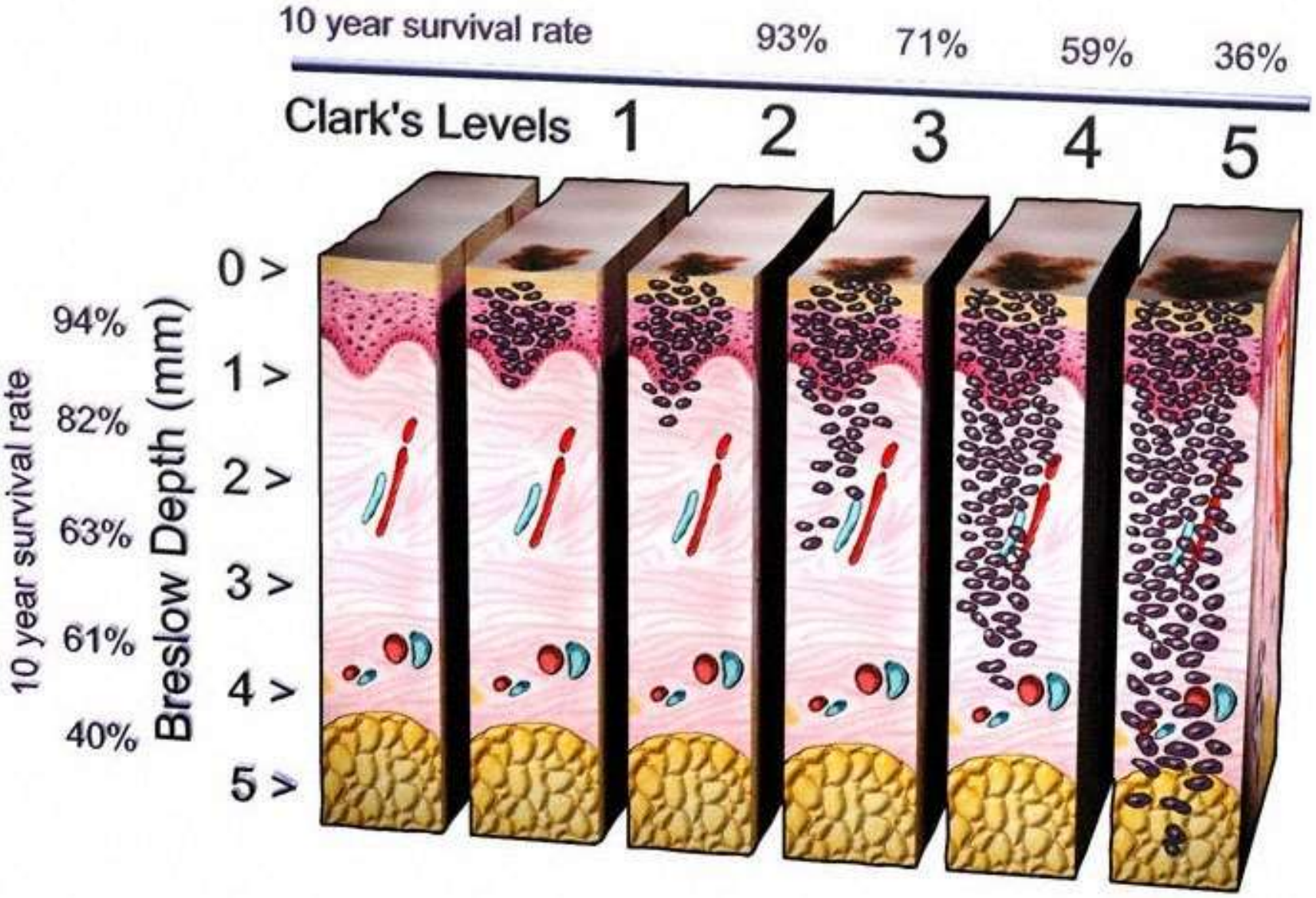
**Clark's level – less accurate**

- Level I: epidermis; called "in situ" melanoma; 100% cure
- Level II: invasion of the papillary (upper) dermis
- Level III: filling papillary dermis, not reticular (lower) dermis
- Level IV: invasion of reticular dermis → ~50% dead in 10 yr
- Level V: invasion of the deep, subcutaneous tissue

**Breslow depth – more accurate**

- 0.75 mm deep → 0% metastases
- 0.85 mm deep → 97% survival in 10 years
- 3.6 mm deep → 50% dead in 10 years




10 year survival rate
93%
71%
59%
36%
Clark's Levels
1
2
3
4
5
10 year survival rate
94%
82%
63%
61%
40%
Breslow Depth (mm)
0 >
1 >
2 >
3 >
4 >
5 >

# Malignant Melanoma  Assessment

- R/O pigmented lesions

Plan:

- Refer for biopsy



# Malignant Melanoma



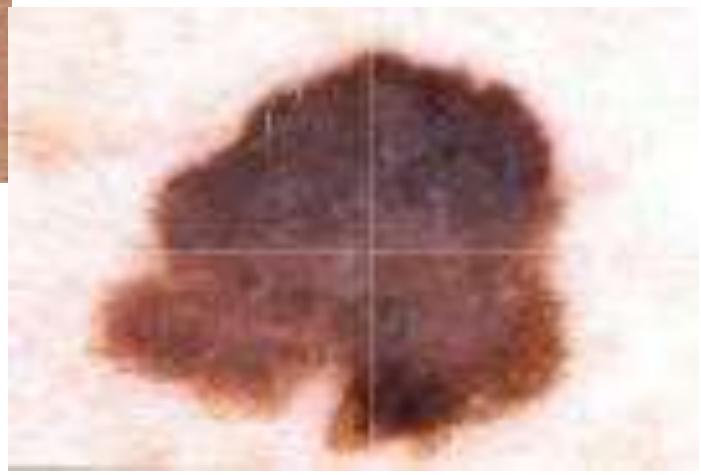
# Malignant Melanoma



# Malignant Melanoma









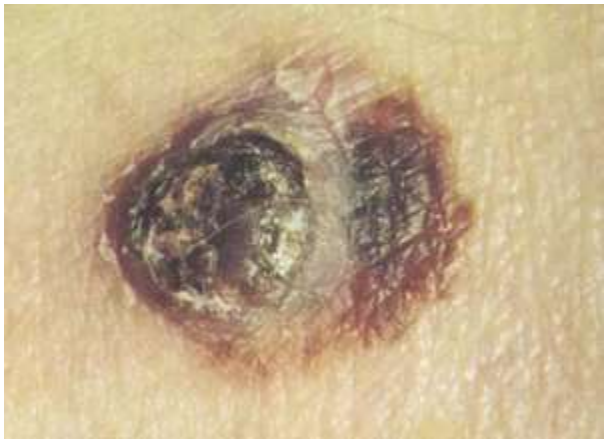
# Superficial Spreading Melanoma





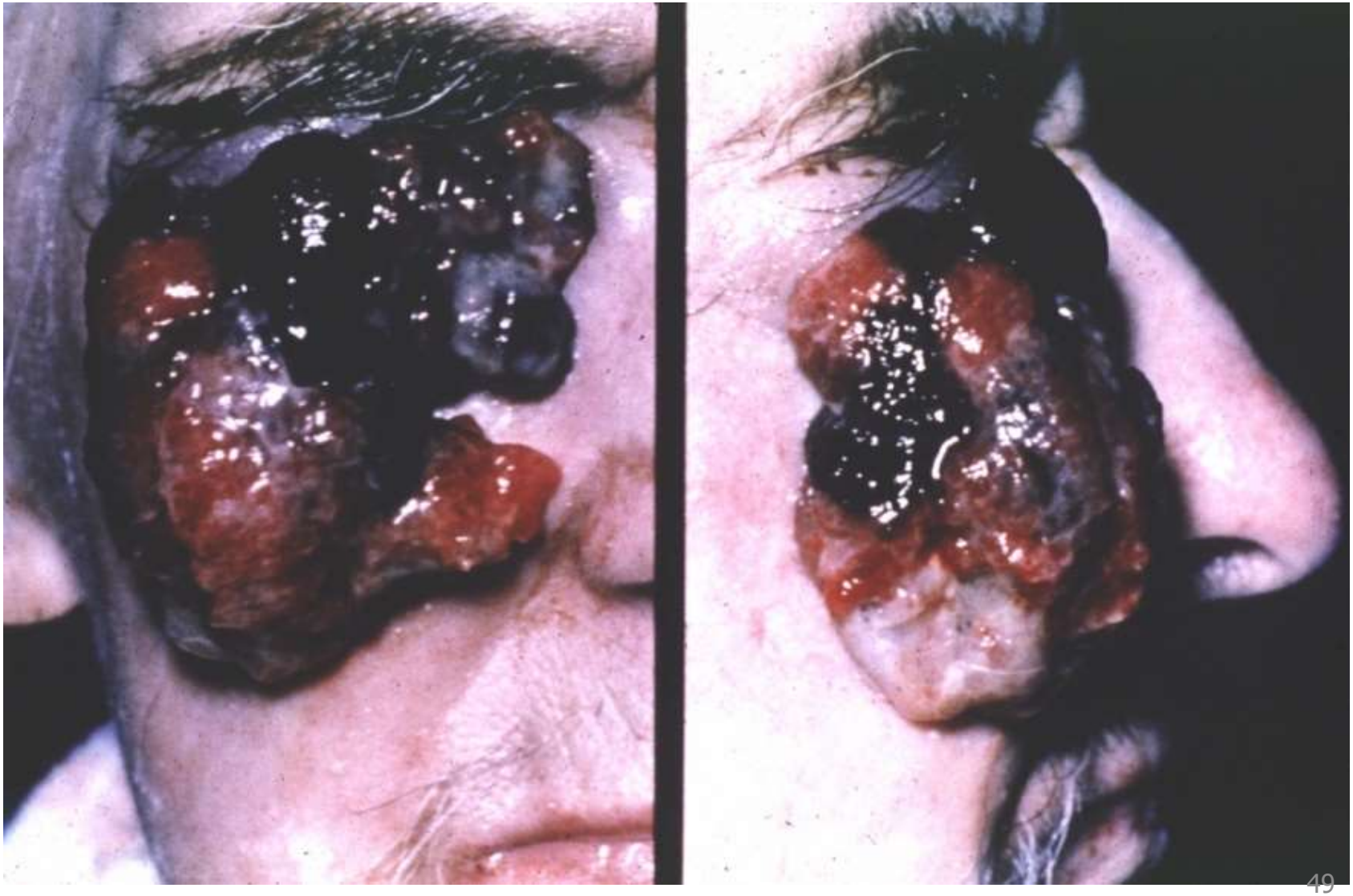

# Nodular Melanoma



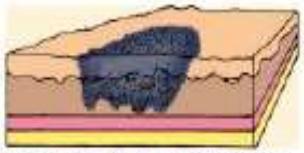


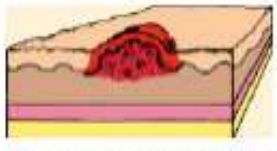


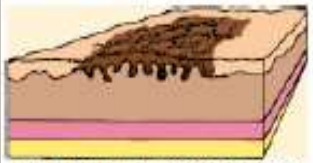
# Malignant Melanoma



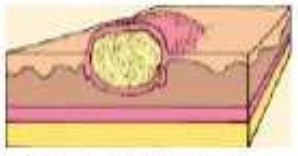
# Eyelid Lesions Len V Koh OD & Lorne Yudcovitch OD

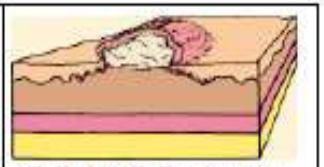

**Macule**: flat, small, multicolor


**Papule**: small(<5mm)elevation

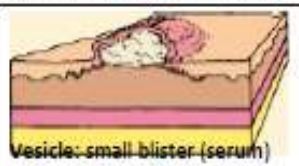

**Plaque**: large(>5mm)flat lesion


**Nodule**: large solid elevation
**Tumor**: large nodule


**Pustule**: purulent elevation

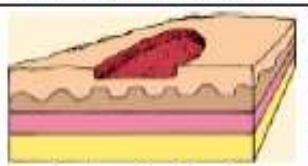

**Wheal**: edematous skin area

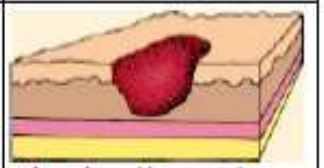

**Cyst**: enclosed cavity with lining (semisolid/liquid)

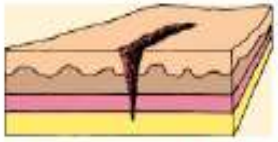

**Vesicle**: small blister (serum)
**Bullae**: large vesicle (>5mm)


**Scale**:dead epidermal cells


**Crust**: dried exudate 'scab'


**Erosion**:loss of epidermis


**Ulcer**: dermal loss, scarring


**Fissure**: deep dermal skin split


Flat-topped (lichenoid)


Dome-shaped


Pedunculated


Filliform


Verrucous


Umbilicated


Erythema: skin redness


Excoriation





Keloid


Lichenification


Sessile squamous papilloma


Pedunculated papilloma


Seborrheic keratosis


Keratoacanthoma


Cutaneous horn


Actinic keratosis


Sebaceous nevus


Basal cell carcinoma


Hemorrhagic BCC


Morpheaform BCC


Squamous cell carcinoma


Sebaceous carcinoma

References: 1. Shields JA; Shields CL. Eyelid, Conjunctival, and Orbital Tumors: An Atlas and Textbook. Philadelphia: Wolters Kluwer Health, 2016. v. 3rd edition
2. Bolognia JL, Schaffer JV, Karynne DO, Ko CJ. Dermatology Essentials. Elsevier Saunders Inc 2014. v. 1st edition

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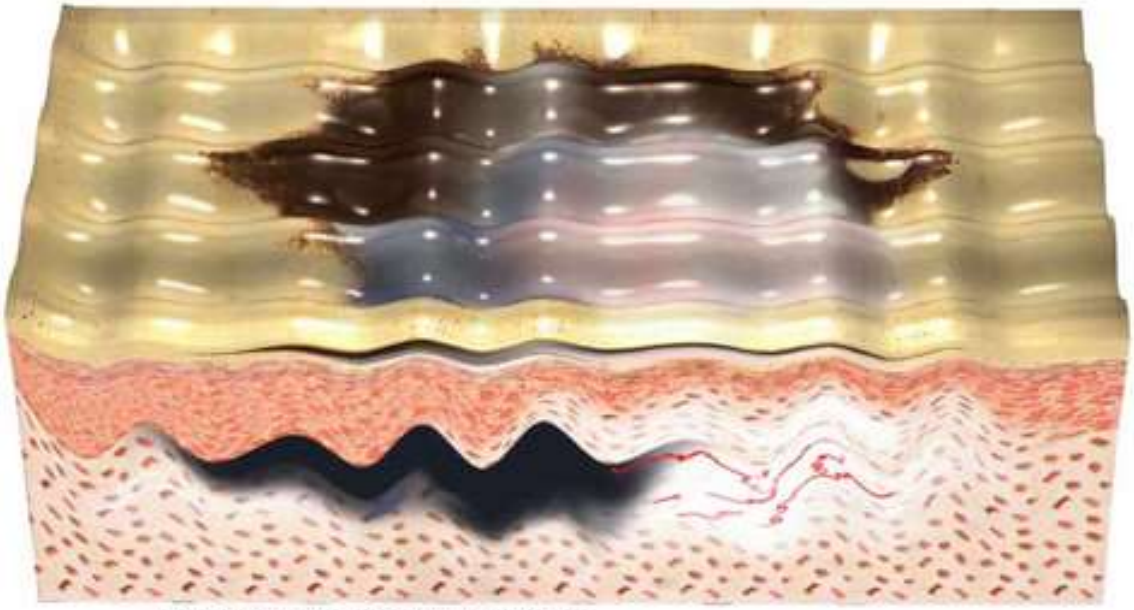
# Epiluminescence Microscopy (ELM)

- Typically x10 mag
- Oil medium between instrument and skin
- Allows inspection unobstructed by skin surface reflections
- Makes subsurface structures viewable in vivo
- New versions use polarized light to cancel reflections





# Epiluminescence Microscopy (ELM)






- Allows magnified visualization of pigmented skin lesions beyond what is visible in normal clinical exam
- Designed for small equivocal pigmented skin lesions
- Also called: dermatoscopy, dermoscopy, skin surface microscopy, incidence light microscopy

# Punch Biopsy





# Punch Biopsy



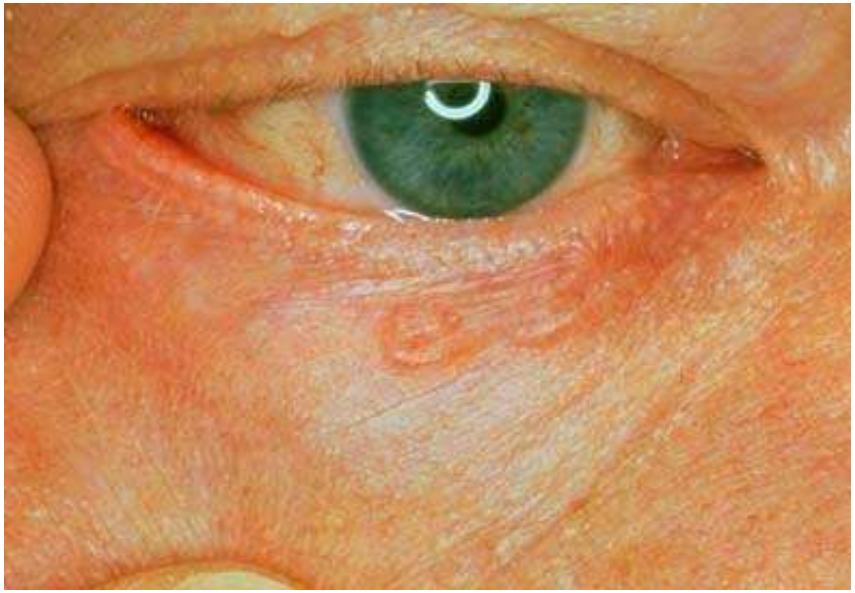
# Mohs Surgery

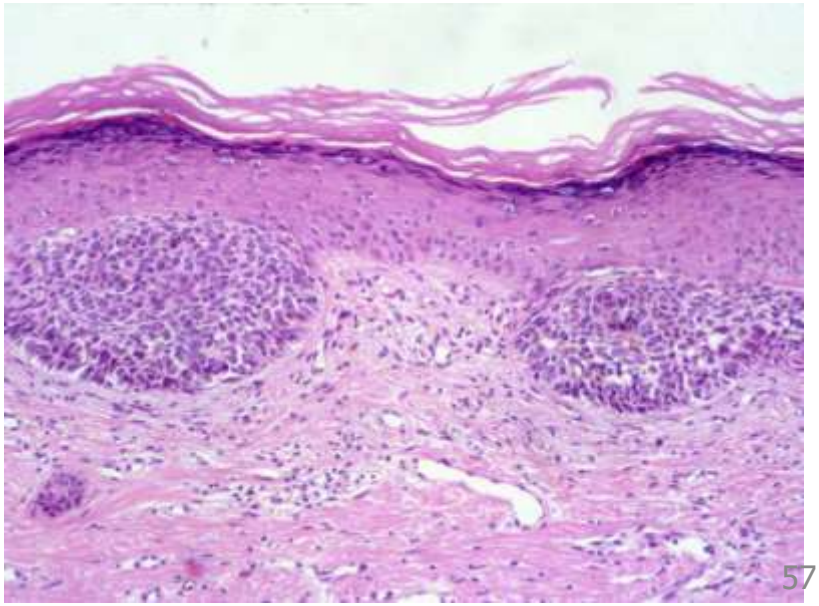


# BASAL CELL CARCINOMA

# Basal Cell Carcinoma

- Malignant tumor
- MOST COMMON MALIGNANT LID TUMOR
- Etiology – UV exposure connection, elderly, fair skin (rare in darkly-pigmented people)





# Early Basal Cell Carcinoma



# Early Basal Cell Carcinoma





# Basal Cell Carcinoma

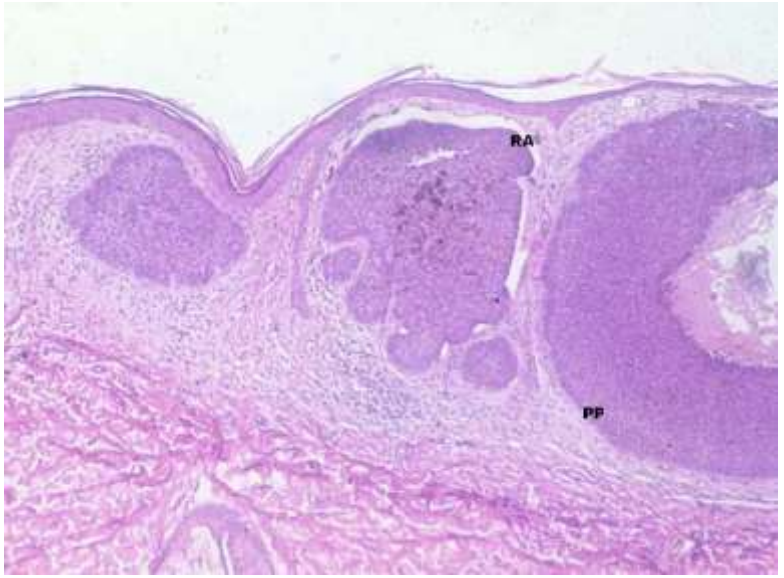


# Basal Cell Carcinoma

Objective:

- Location
- Types
  1. Sclerosing
  2. Noduloulcerative



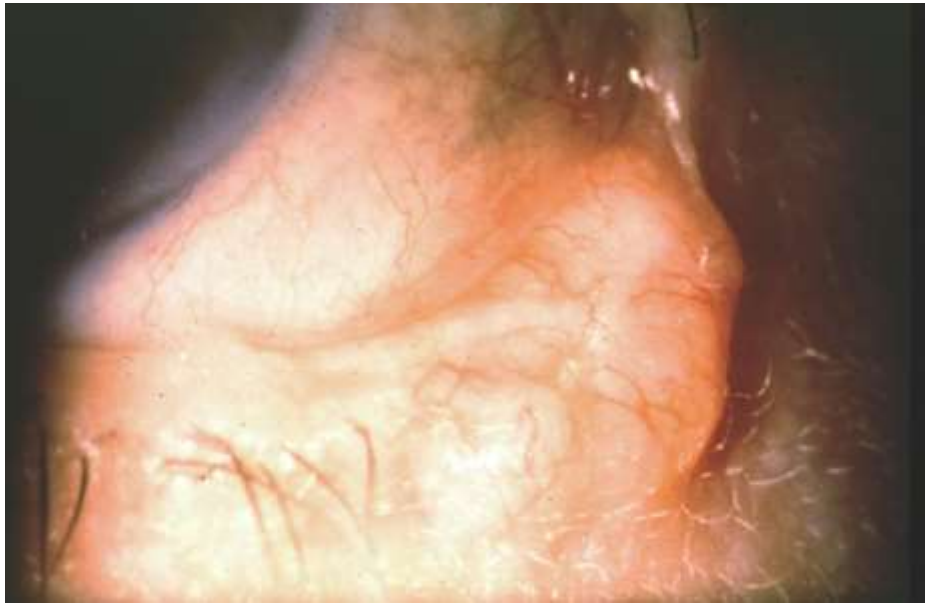



- Early forms look like vascularized nodules
- Varying degrees of central umbilicated ulceration
- ”Pearly”, rolled borders
- Pigmentation
- Surface may become inflamed, infected or both

# Ulcerative Basal Cell Carcinoma



# Basal Cell Carcinoma
# Nodular Ulcerative (left) Sclerosing Type (right)





# Basal Cell Carcinoma
# Sclerosing Type

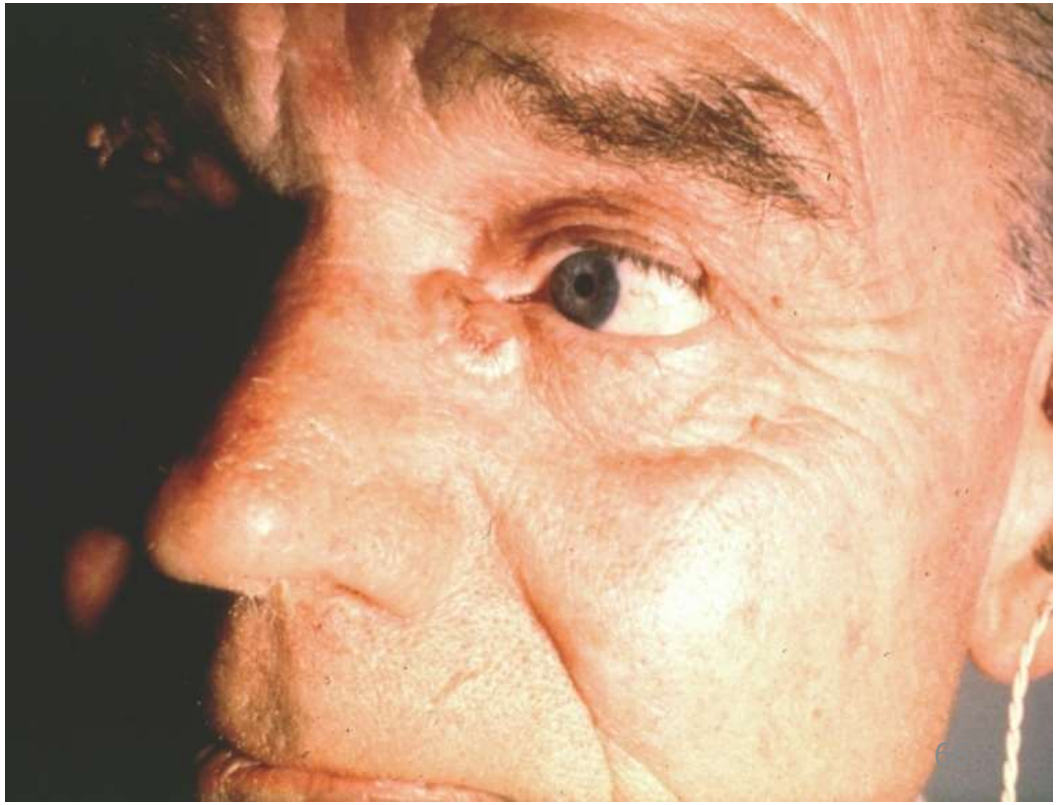


# Basal Cell Carcinoma Nodulo-Ulcerative



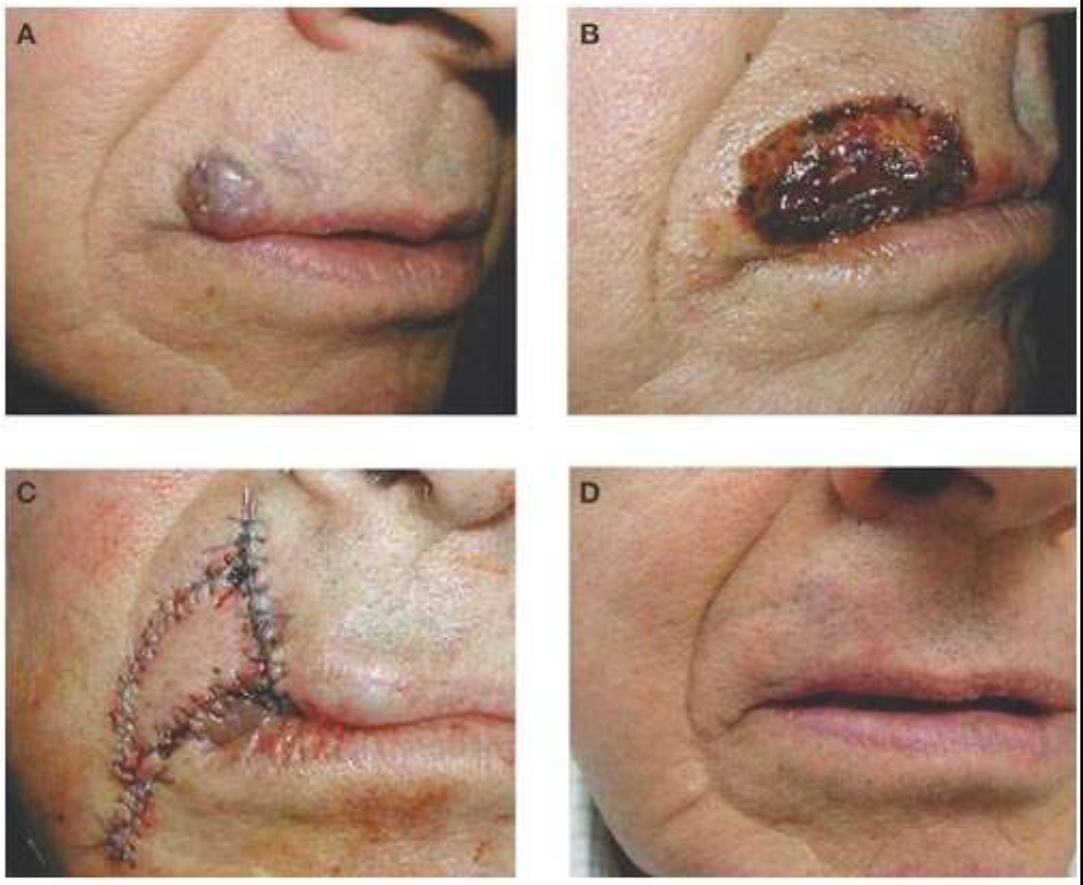
# Basal Cell Carcinoma Assessment/DDX

- Metastasis uncommon
- Inner canthal lesions require more rapid attention
- Consider all lid lumps & bumps!
  - Keratoacanthoma
  - HSV lesion

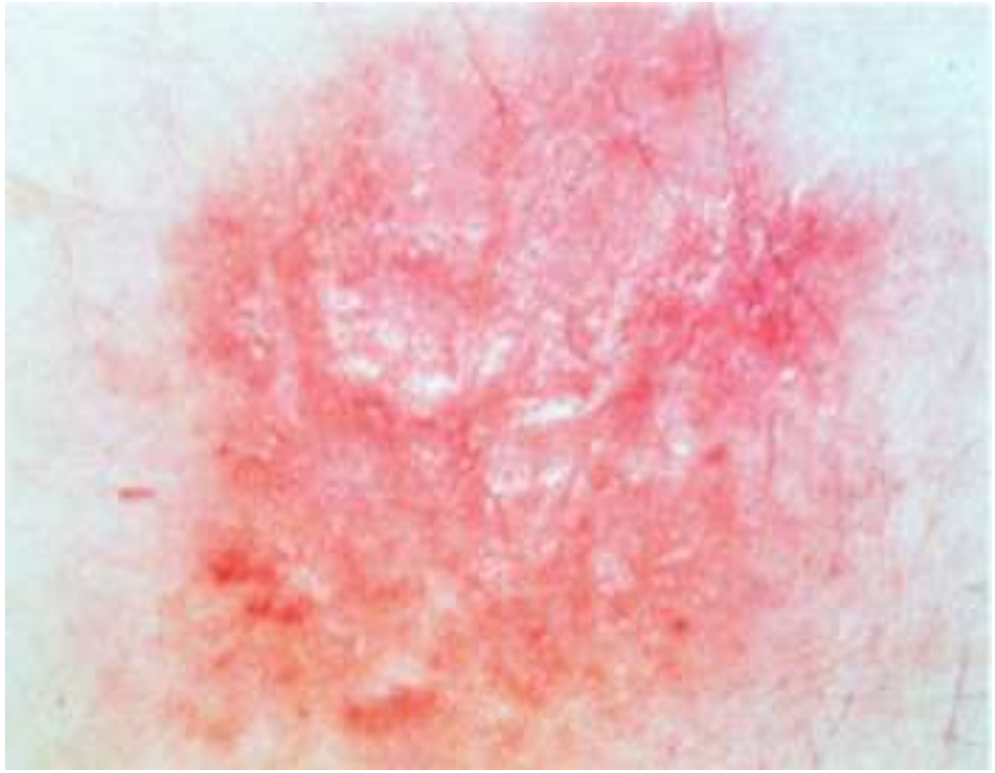


# Basal Cell Carcinoma Plan/TX

- If secondarily infected or inflamed, run short course of AB-steroid combo
- Derm. consult
- surgical excision & repair
- radiation
- cryotherapy
- curettage
- electrodessication
- Imiquimod cream (Aldara)
- 5-FU (fluorourocil)
- PDT (photodynamic therapy)
- Pt. education



# Basal Cell Carcinoma

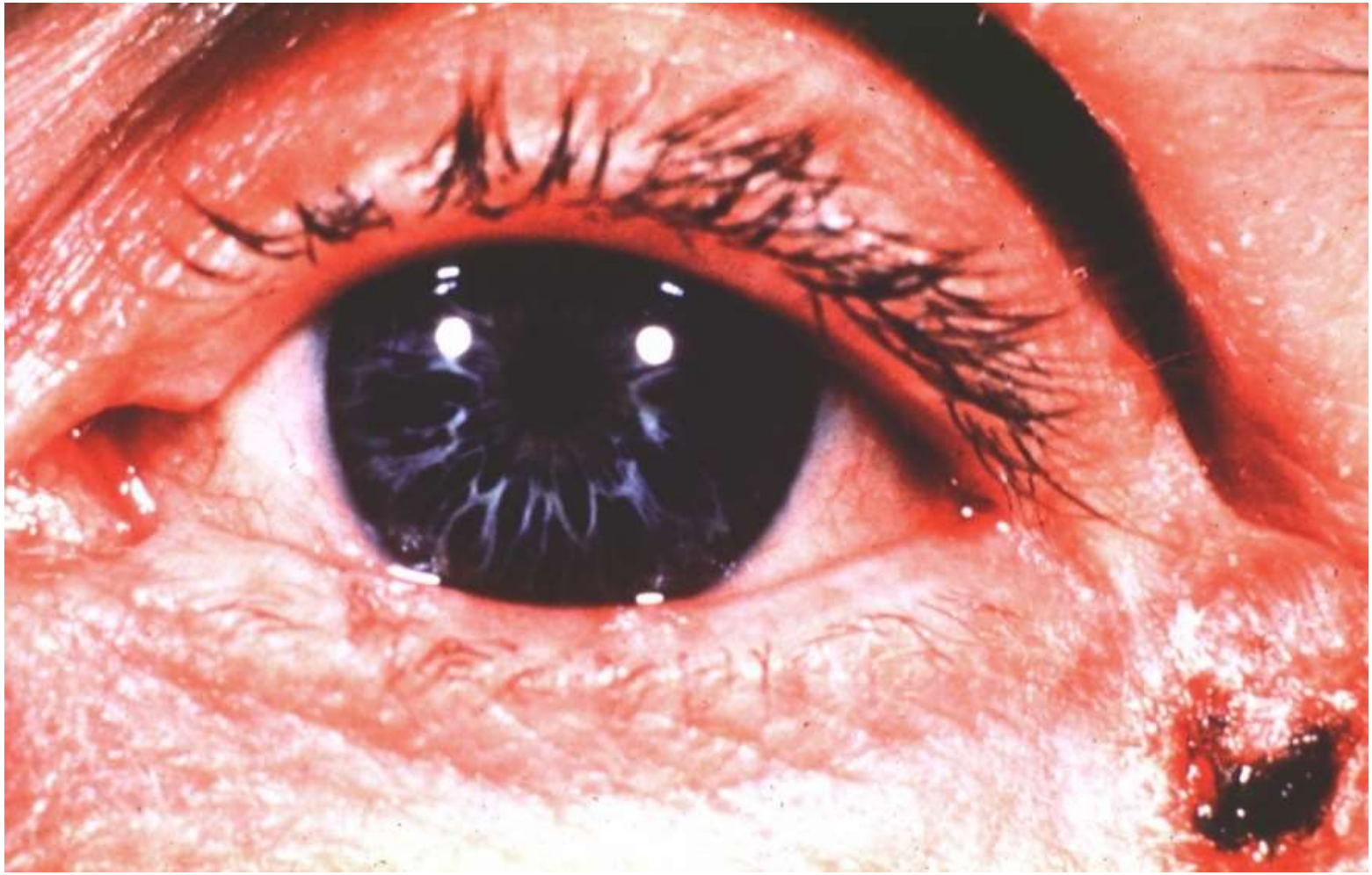




# Basal Cell Carcinoma



# Basal Cell Carcinoma Infected and Inflamed



# Basal Cell Carcinoma Infected and Inflamed



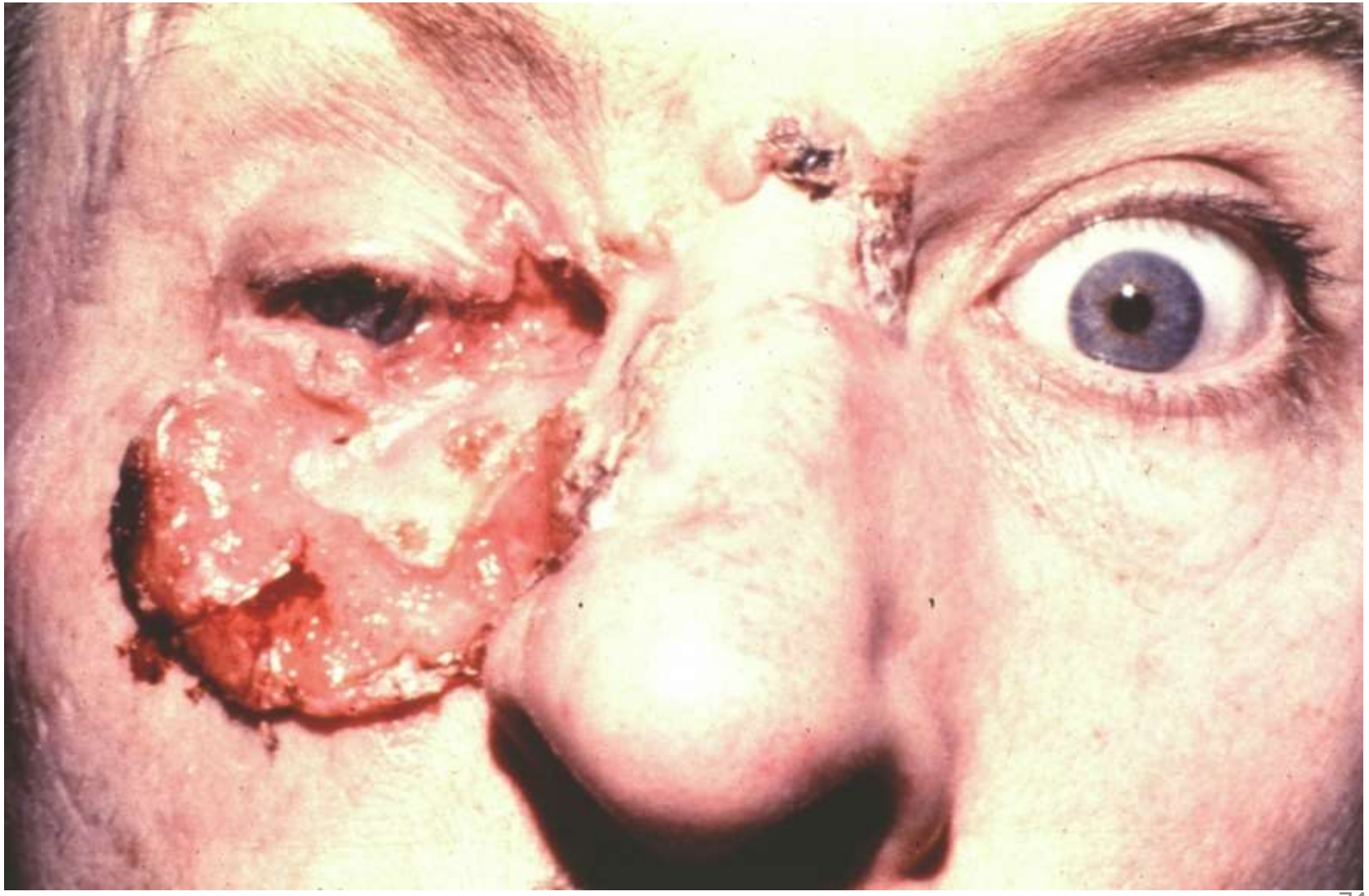
# Basal Cell Carcinoma



# Basal Cell Carcinoma



# Basal Cell Carcinoma

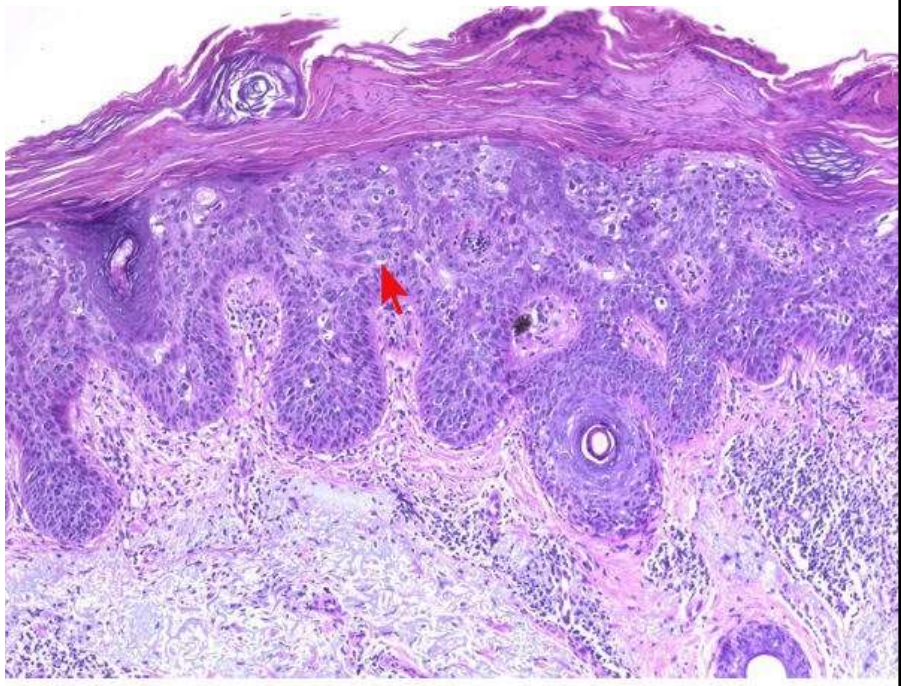


# SQUAMOUS CELL CARCINOMA

# Squamous Cell Carcinoma

- Malignant tumor
- 3RD MOST COMMON OCULAR TUMOR
- 2nd most common skin cancer
- Etiology: Keratinizing epidermal cells





Intraepidermal atypical squamous cells

# Squamous Cell Carcinoma

- Subjective – scab that doesn't heal
- Objective - reddish, raised scaly plaque
- Assessment/DDX - R/O all lid lumps & bumps!
  - Keratoacanthoma
  - sebborheic keratoses
  - cutaneous warts
- Plan/TX - excision, radiation, PDT (photo-dynamic therapy)



# Squamous Cell Carcinoma



# Squamous Cell Carcinoma



# Squamous Cell Carcinoma





# MEIBOMIAN GLAND CARCINOMA

# Meibomian Gland Carcinoma

- Mistaken for persistent, recurrent meibomian gland chalazion
- A type of sebaceous gland carcinoma
- Can mimic a chronic eyelid/conjunctival infection
- Plan/TX
  - Excision
  - Radiation



# Meibomian Gland Carcinoma



# Meibomian Gland Carcinoma



# PRECURSORS TO CANCER




Lone Pine Koala Sanctuary,
Brisbane www.koala.net





# Actinic Keratosis

- = solar keratosis = “sun spots”
- Squamous cell dysplasia

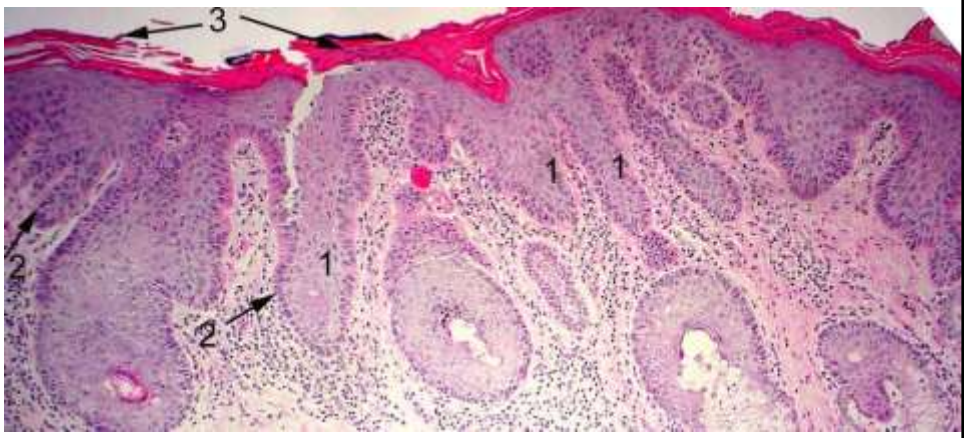
Etiology – UV exposure

Subjective

- Pt unaware
- Pt notices ‘skin spots’

Objective:

- Dry, scaly lesions 2-5 mm
- Minimally elevated






# Actinic Keratosis

## Assessment:

- R/O neoplasia
- R/O keratoacanthoma
- R/O melanoma, nevi, verrucae, papilloma
- R/O seborrheic keratoses



## Plan/TX:

- Dermatology consult
- Cryo, cutterage, topical anti-cancer cream (e.g. imiquimod), PDT

# Actinic Keratosis



# Actinic Keratosis



# Actinic Keratosis



# Keratoacanthoma

- Papular lesion originating from sebaceous glands
- Benign proliferation of squamous epidermal cells

Etiology – unknown (possibly UV, trauma, age)

Subjective – fast growing





A



B

# Keratoacanthoma

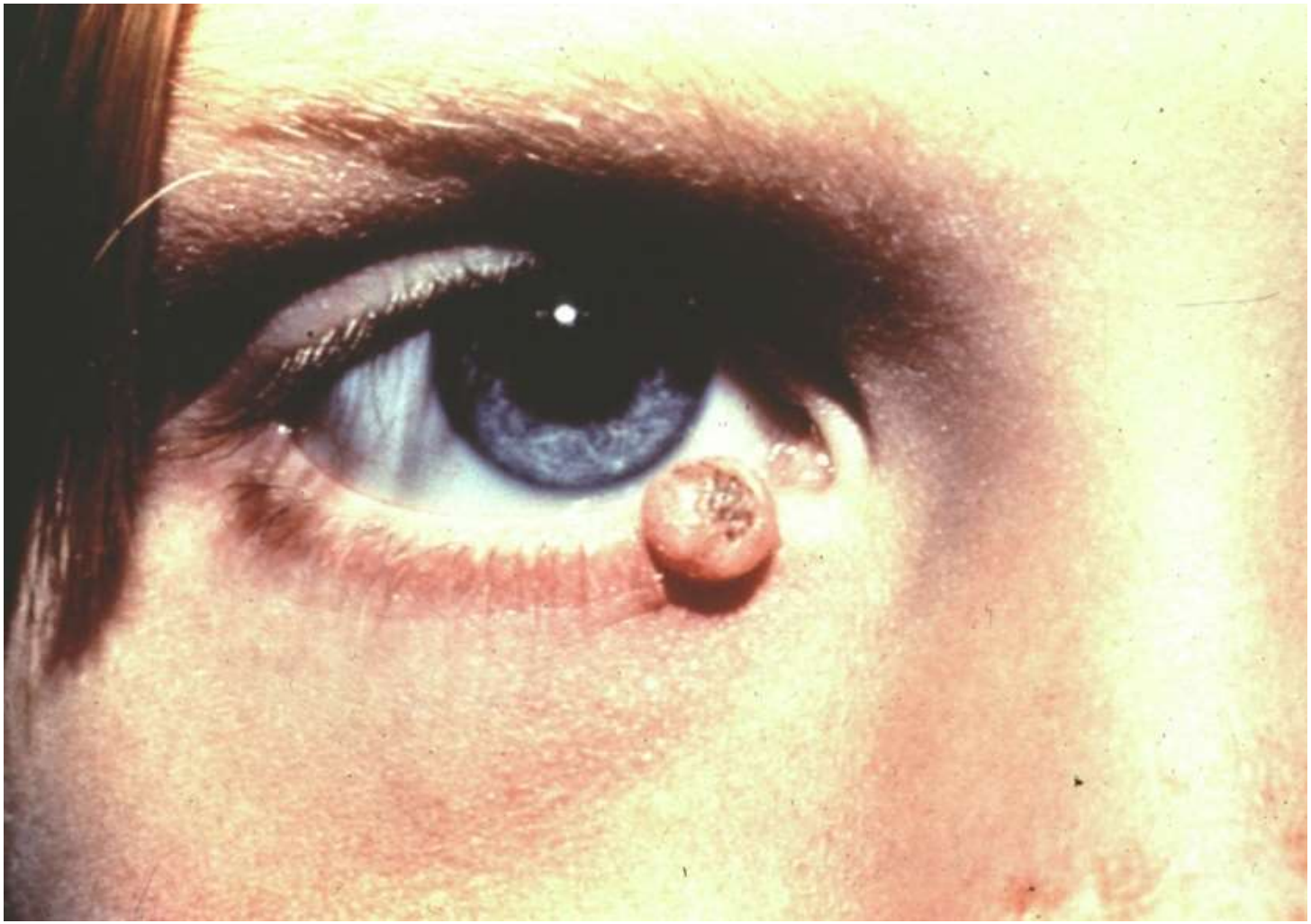
Objective

1. Starts as erythematous papule
2. Grows into firm, raised indurated nodule
   - up to 2 cm
3. Keratin filled crater
4. Color
5. Spontaneously regresses



- Assessment - R/O SCC
- Plan/TX - Derm. consult, excision

# Keratoacanthoma



# Keratoacanthoma



# Keratoacanthoma



# Keratoacanthoma



# Lentigo maligna (Hutchinson's freckle)

- A sub-type of melanoma *in situ*
- Associated with sun exposure (Australia, Hawaii)
- Above age 40; mean age 65.  Fair-skin association
- Primarily in head and neck
- Flat pigmented patches
- 15% malignancy risk



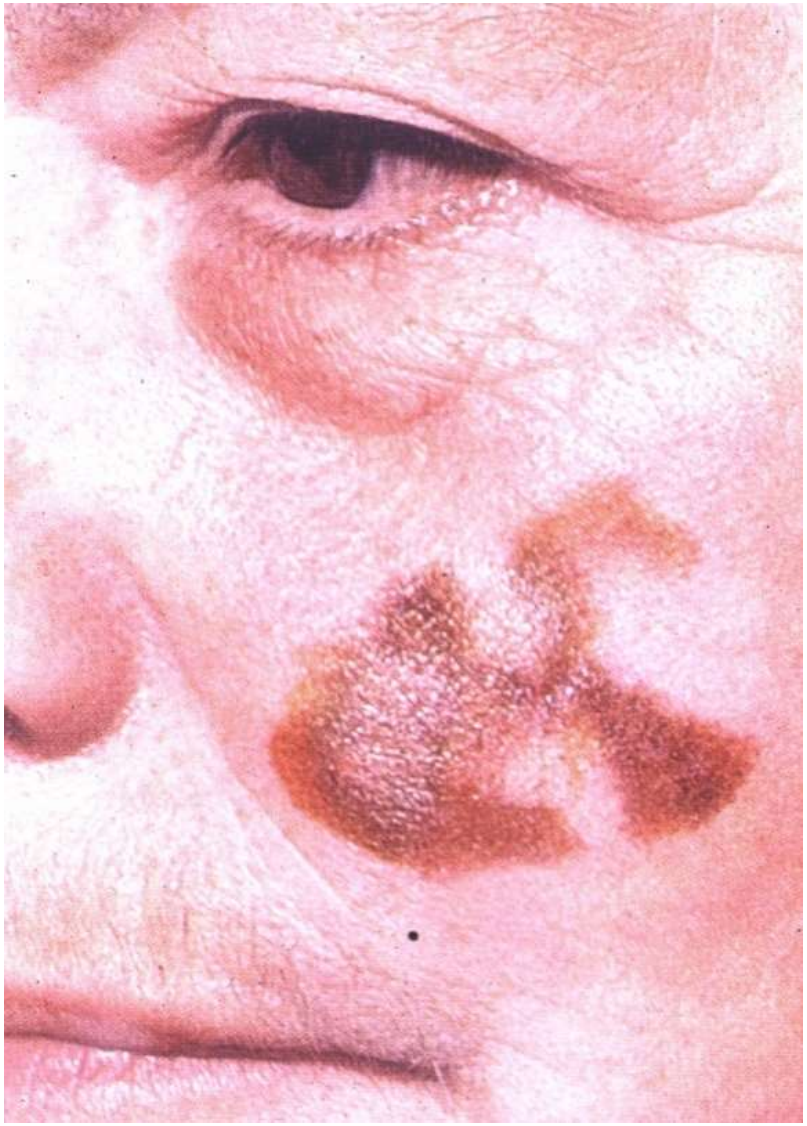


# Lentigo maligna (Hutchinson's freckle)

- Objective - dark macular lesion with irregular serpiginous pigment and margins
- Assessment - R/O other pigmented lesions
- Plan - Refer for biopsy – surgical removal



# Lentigo maligna (Hutchinson's freckle)



# Lentigo maligna (Hutchinson's freckle)

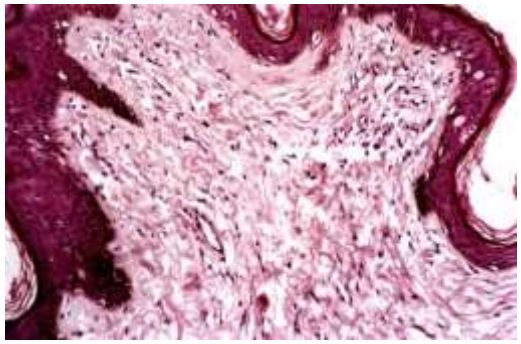


# BENIGN SKIN LESIONS

# Acrochordon



- = “skin tag”, soft warts
- Incidence 46%
- small, soft, common, benign, usually pedunculated neoplasm found particularly in the obese
- Etiology - Localized paucity of elastic tissue



Mgt:

- Snip it
- Freeze it
- Zap it
- Chromium picolinate 200 mcg BID?



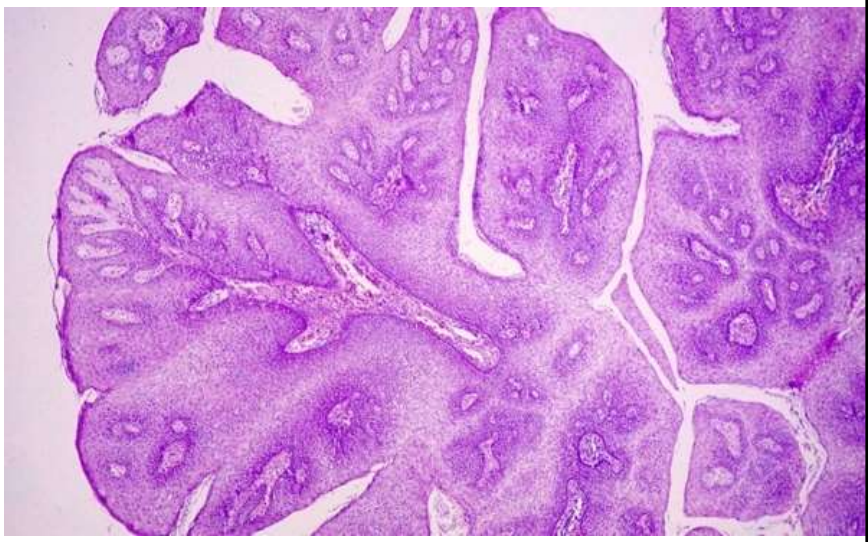
# Acrochordon (Skin Tag)





# Papilloma

- Benign epithelial tumor
- Etiology – epithelial overgrowth
- Lobulated in appearance







# Papilloma - Objective

- Size and shape
- Pigmentation
- Number
- Location
- Avascular
- Growth
- Surface is roughened but not eroded



# Papilloma - Assessment

R/O other lumps & bumps:

- Nevus
- Keratoses
- Seborrheic cyst
- Verrucae
- Molluscum
- Squamous cell carcinoma
- Other neoplasia





# Papilloma Plan/TX

- Reassurance and pt. education
- Removal
- Derm. consult
- RTC q. 3-6 mos.
- RTC q. 1 yr.





- Papillomae can be sessile (flattened) or pedunculated (elongated)
- They can be the same color as skin or pigmented
- They often look like cauliflower!

# Papilloma

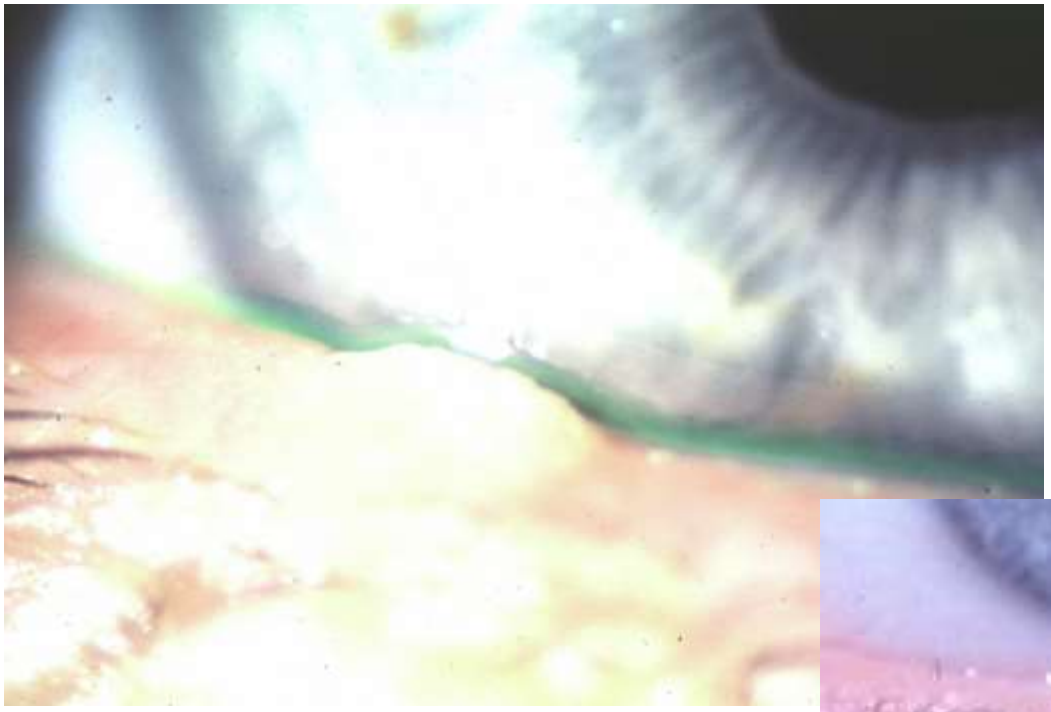


# Pigmented Papilloma





# Papilloma

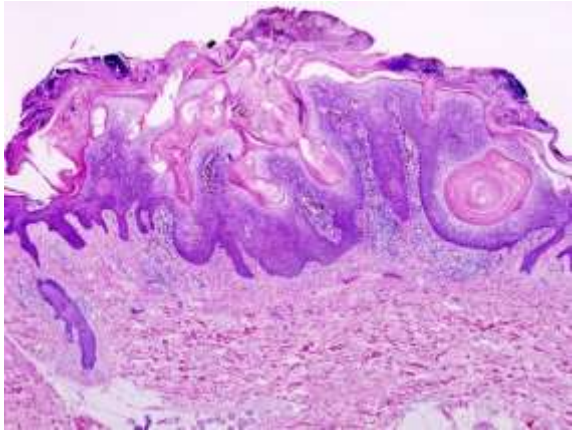




# Seborrheic keratosis



- = “barnacles of old age”
- Etiology – basal cell hyperplasia
- Hyperkeratinized plaques
- Most common benign epidermal tumor in the elderly



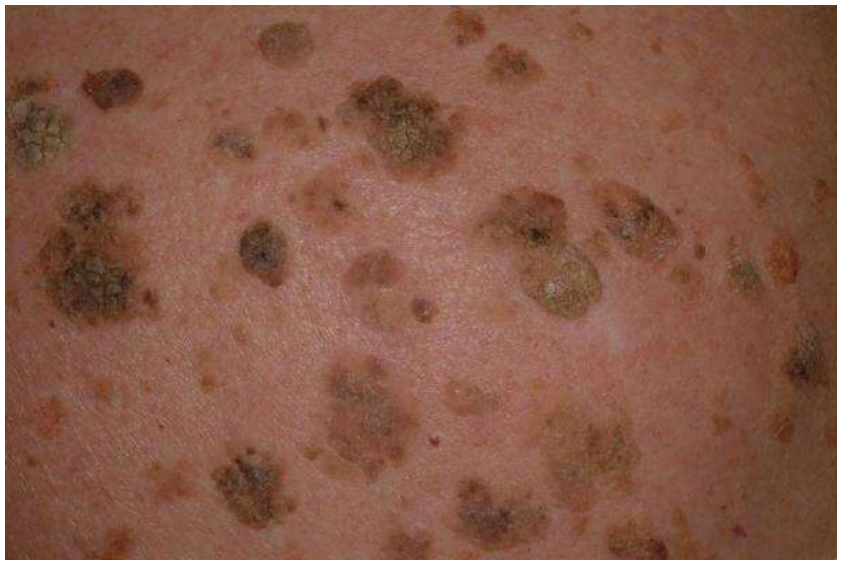




# Seborrheic keratosis

Objective:

- Flat or slightly elevated 1-6 mm round to oval
- Waxy, greasy or scaly lesion





# Seborrheic Keratosis Assessment

Rule-out:

- neoplasia
- keratoacanthoma
- melanomas, nevi, verrucae, papilloma, actinic keratosis
- Plan/TX
  - Derm. Consult
  - cryo, excision, cuterrage
  - RTC q. 6-12 mos.

| Normal Mole | Melanoma | Sign | Characteristic |
|---|---|---|---|
|  |  | Asymmetry | when half of the mole does not match the other half |
|  |  | Border | when the border (edges) of the mole are ragged or irregular |
|  |  | Color | when the color of the mole varies throughout |
|  |  | Diameter | if the mole's diameter is larger than a pencil's eraser |

# Seborrheic Keratosis



# Seborrheic Keratosis



# Seborrheic Keratosis



# Seborrheic Keratosis



# Lichen planus

- Common pruritic eruption affecting pts between 30-60 yo
- Reported in approximately 1% of all new pts seen at dermatology clinics
- Flat topped violaceous papules (2-4 mm)
- Etiology - cell mediated immune response of unknown origin





# Lichen planus

Also commonly affects buccal (mouth) mucosa

Management:

- Biopsy prudent
- Steroids
- Vitamin A
- Retinoids



# Lichen planus



# Lichen planus

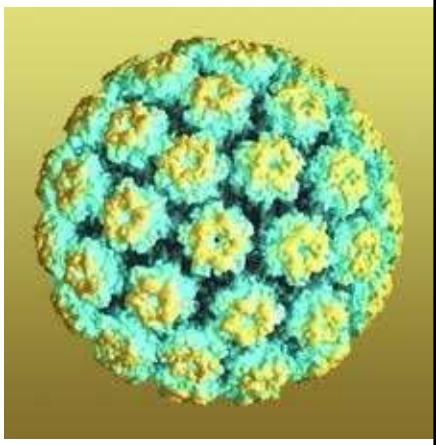




# VIRAL SKIN LESIONS

# Verrucae

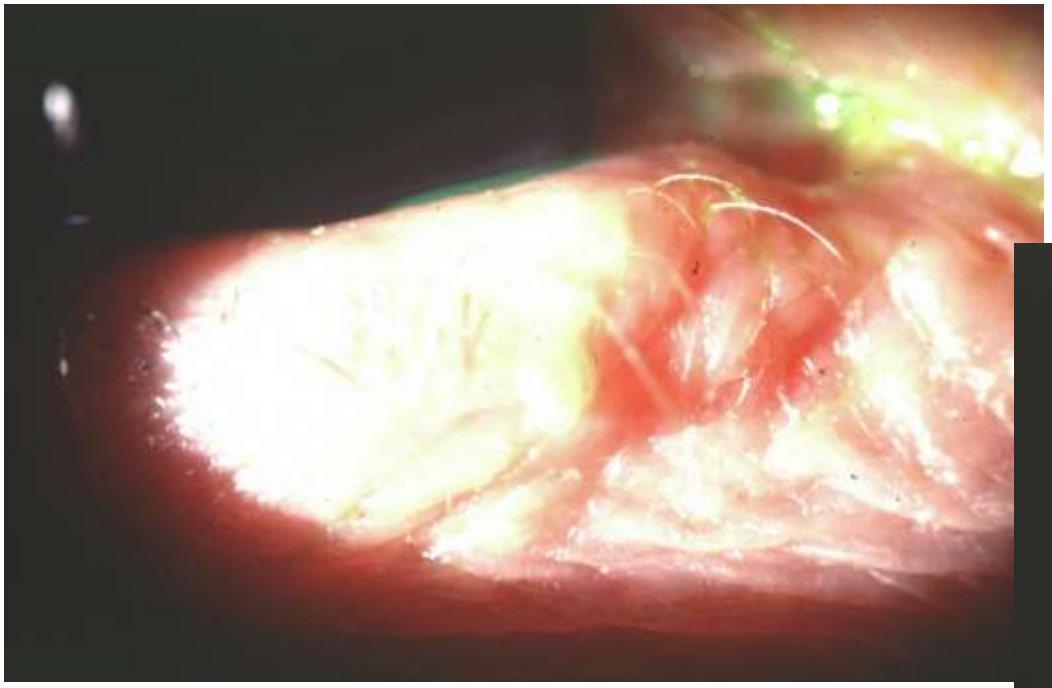
- = viral warts
- Etiology - Virus (HPV – Human Papilloma Virus)
- Plantar wart (verruca plantaris) – wart on sole or toes of foot
- Benign skin tumor

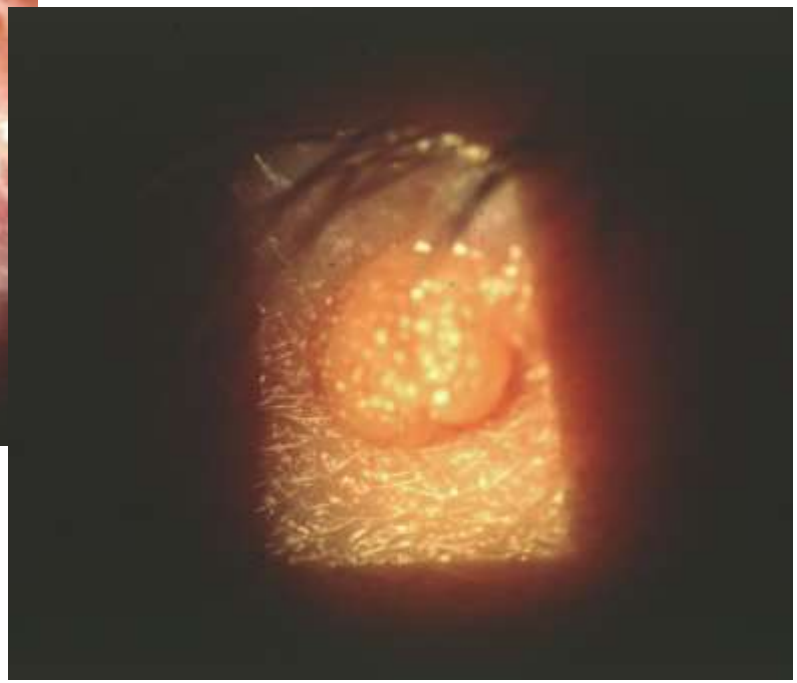






# Verrucae





# Verrucae





# Verrucae

Objective

- Single or multiple non-secreting warts
- Color
- Various shapes:
  1. Verruca plana
  2. Verruca vulgaris
  3. Verruca digitata

Assessment/DDX

- R/O other lumps & bumps – papilloma, molluscum contagiosum

# Verrucae Plana





# Verrucae Vulgaris

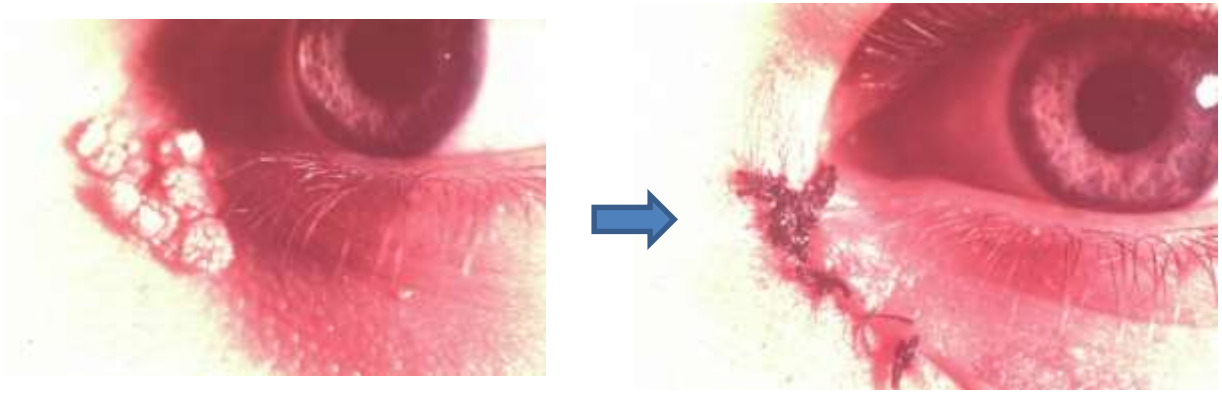


# Verrucae Digitata



# Verrucae Plan/TX

- Resolution
- Reassurance and patient education
  - Contagious on skin contact, dressing room floors
- Removal
  - Excision
  - Keratolytic agents (dichloroacetic acid)
  - Laser
  - Cryo
  - Duct tape?



Multiple verruccae removed with excision

# Verrucae



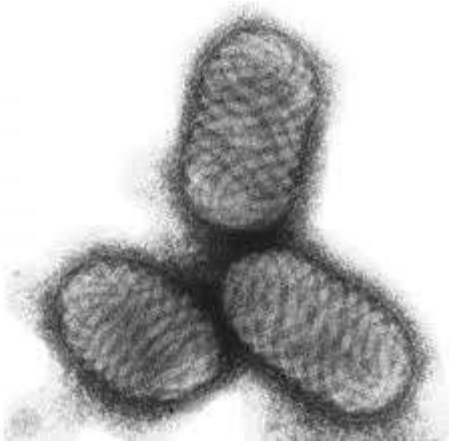
# Verrucae





# Molluscum contagiosum

- Benign skin tumor
- Etiology - Virus (pox), association with STD and HIV
- Contagious (skin-skin, swimming pools, etc.)
- Children can pass it between them frequently





# Molluscum contagiosum

- Almost exclusively human disease
- Member of the Poxvirus family; DNA virus
- Four main types (I-IV)
  - Type I: over 95% of infections
  - Type II: in 60% of HIV infections
  - Types III and IV: very rare
- Inoculation via skin transmission
  - Person to person
  - Autoinoculation

# Molluscum contagiosum

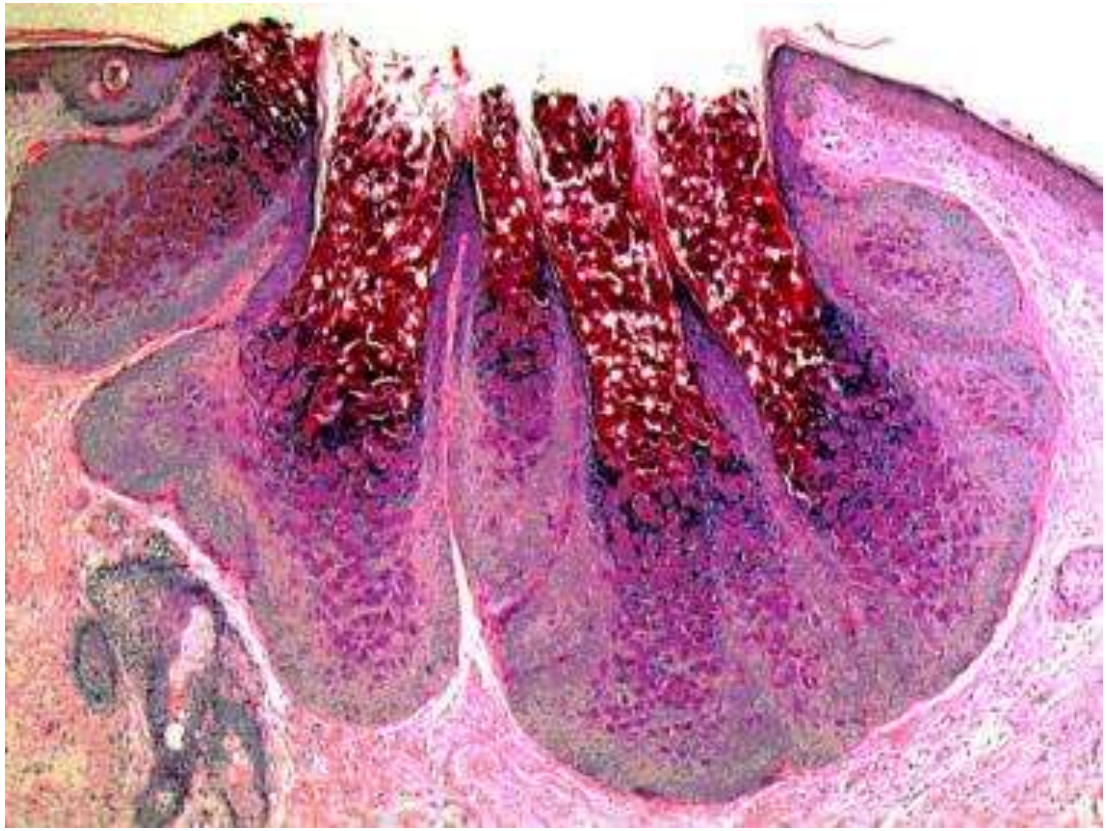
Objective:

- Umbilicated nodule with central core
- Elevated, round, waxy, pearly
- Follicular conjunctivitis
- Multiple
- Keratitis





Lesion crush/squash preparations can confirm Henderson-Paterson bodies (intracytoplasmic virions)



Viral incubation and replication leads to epithelial cellular proliferation with umbilicated core



# Molluscum contagiosum



Assessment:

- DDX – R/O BCC

Plan/TX:

- If quiet, leave alone
- If central discharge, express
- Excision, cryo, topical acids, oral medication
- Home remedies (variable benefit)
- Pt. ed. – cleanliness, transmission
- If large number of lesions (100+), consider HIV

# Molluscum contagiosum

**Topical treatments:**

- Podophyllotoxin cream (0.5%)
  - Fetus toxicity; avoid with pregnancy
- Iodine
- Salicylic acid (ASA, Aspirin)
- Potassium hydroxide
- Tretinoin cream
- Cantharidin (blistering agent)
- Imiquimod (T cell modifier)
  - not recommended for children due to possible adverse events and not effective in children




**Oral treatment:**

- Cimetidine



# HOME REMEDIES FOR
# MOLLUSCUM CONTAGIOSUM



Apple Cider Vinegar




Tea Tree Oil




Hydrogen Peroxide



Indian Lilac




Australian Lemon Myrtle



Grapefruit Seed Extract



Colloidal Oatmeal




Clove Oil



Oil of Oregano



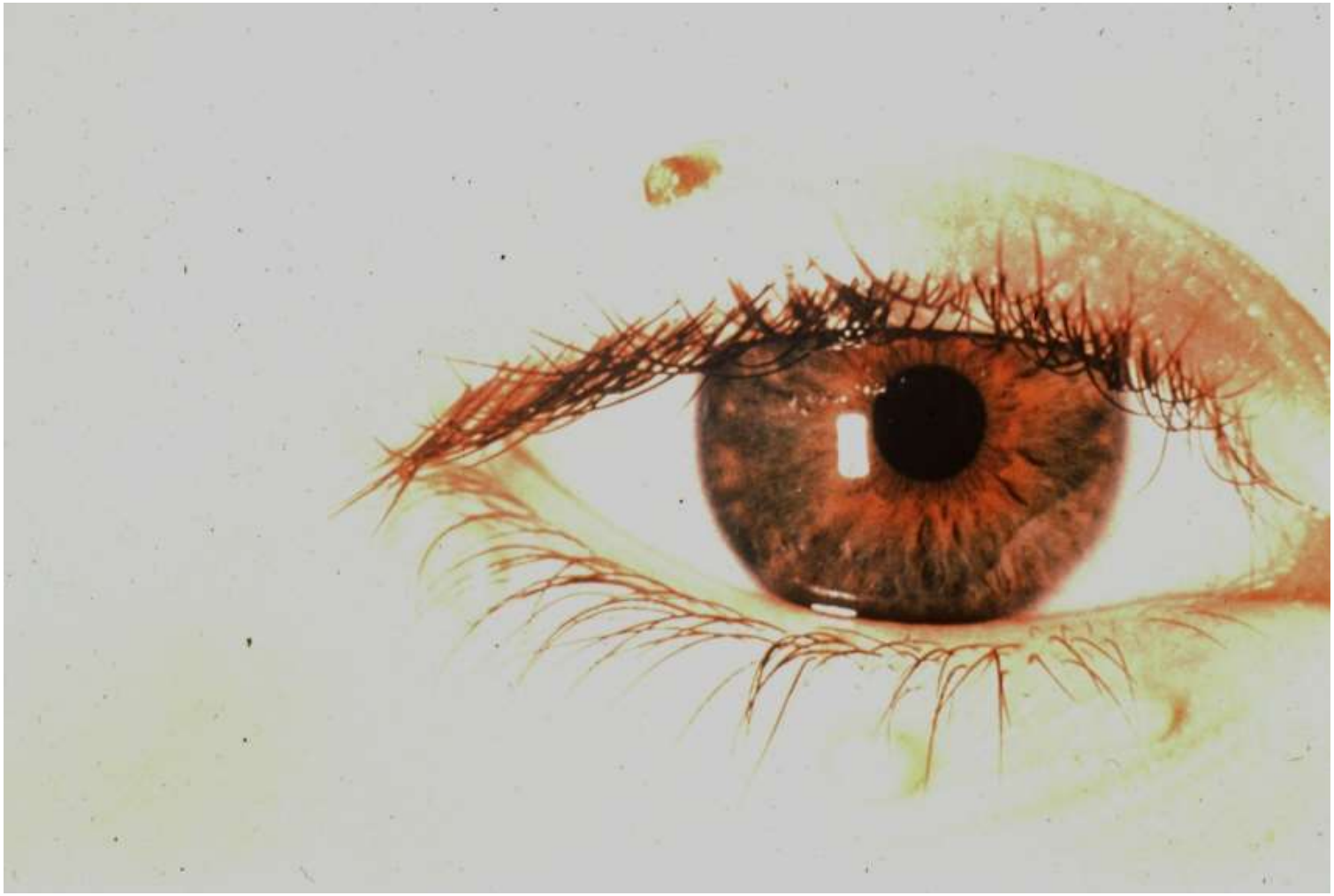
Top 10 Home Remedies

To explore more, visit
www.Top10HomeRemedies.com

# Molluscum contagiosum

- Excellent prognosis; tend to self-resolve (months)
- About a third of patients may get recurrence
- Hand wash, cover if possible, avoid scratching
- Avoid physical contact and sharing towels/clothes
- Adolescents/adults: sexual transmission common

# Molluscum contagiosum



# Molluscum contagiosum



# Molluscum contagiosum





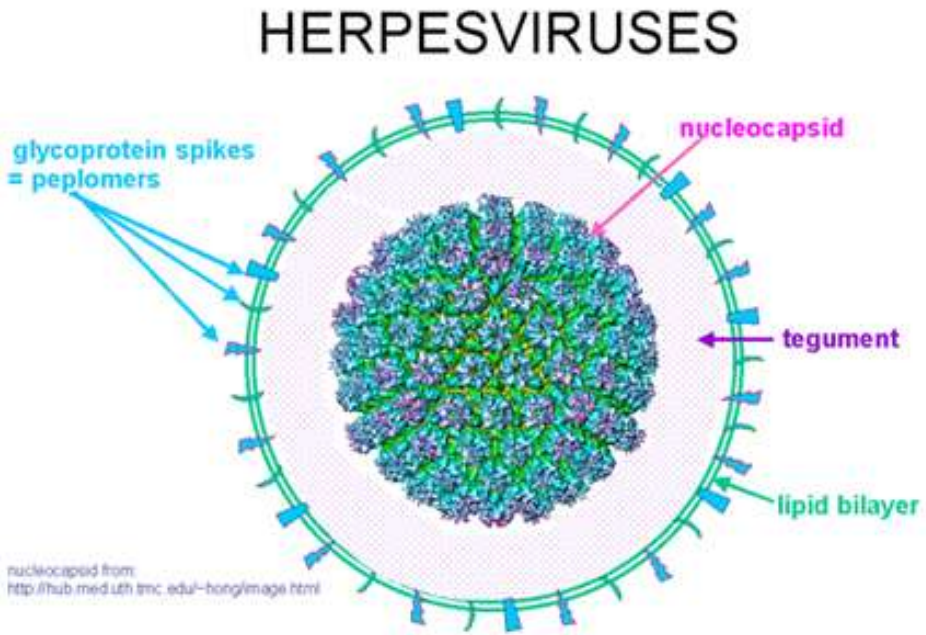
# Molluscum contagiosum



# Herpetic Eyelid Lesions

There are 2 main herpes eye infections:

- Herpes zoster
- Herpes simplex
- Each of these important virions can cause MANY ocular diseases (including eyelid lesions)




- These will be covered in more detail later in the course

# Herpes simplex virus (HSV) primary lesions



# Varicella zoster virus (VZV)
# aka herpes zoster
# Shingles lesions



Hutchinson's sign (nose)

Thoracic dermatome



Shingles

# THANK YOU!